Cyclistic Bike Share: Member vs Casual Rider Behavior

End-to-End Analysis Project

APPASA Strategy Document

Google Data Analytics Capstone

Introduction

I will use this APPASA strategy document to record my decisions and reflections as I work through this capstone project. This document will serve both as a guide to help me think through my responses and reflections at different stages of the data analysis process, and as a valuable resource I can reference in the future to support my growth as a data professional.

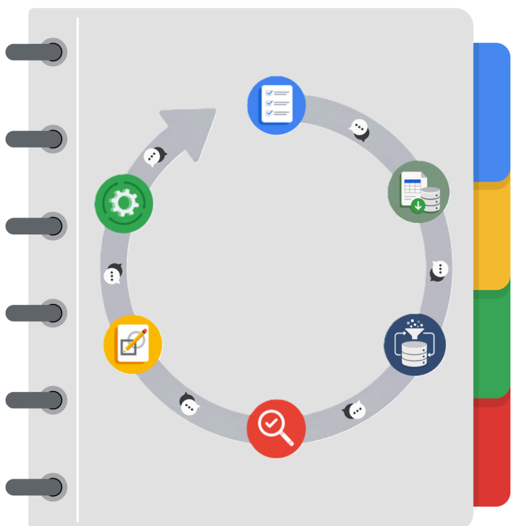
Portfolio Project Recap

Many of the goals I accomplished throughout the individual course portfolio projects are integrated into this end-to-end data analysis capstone project. These include:

- Creating a clear and structured project proposal.
- Demonstrating my understanding of SQL and relational database concepts.
- Using SQL to ingest, explore, clean, transform, validate, and organize data through structured queries and database operations.
- Organizing and analyzing a large-scale dataset to uncover meaningful insights and communicate the underlying business story.
- Developing a series of SQL scripts for ELT, feature engineering, data validation, and analytical querying, alongside a DuckDB-powered Jupyter Notebook documenting the complete APPASA methodology and reproducing the analytical workflow through SQL queries and visualizations.
- Computing descriptive statistics, performing exploratory and business-focused analyses, and creating analytical visualizations to support data interpretation and executive decision-making.
- Communicating findings through an executive summary, an interactive Power BI report, strategic business recommendations, and implementation-focused action plans for stakeholders.
- Translating analytical findings into evidence-based business recommendations, implementation strategies, success metrics, and future analytical opportunities through the final Act stage of the APPASA framework.

This capstone brings together the knowledge and technical skills I developed throughout the program, demonstrating my ability to execute a complete end-to-end data analysis project, from business problem definition to actionable business recommendations.

THE APPASA WORKFLOW



I will demonstrate to the company's marketing analytics team how I would apply the APPASA workflow to the upcoming Cyclistic Bike Share project. For each question presented in this APPASA strategy document, I will provide a structured and thoughtful response aligned with the corresponding stage of the APPASA framework: Ask, Prepare, Process, Analyze, Share and Act. This approach ensures clarity in my methodology, highlights my strategic planning skills, and illustrates a well-organized path from problem understanding to actionable insights.

Project Tasks

The following questions have been identified as essential to the Member vs Casual Rider Behavior Analysis Project. I will address each question by aligning it with the most appropriate stage of the APPASA workflow: Ask, Prepare, Process, Analyze, Share and Act, based on the specific stage of the project workflow to which the question belongs. To ensure each response is informed and well-reasoned, I will draw on relevant materials from the SQL scripts, project notebook, the APPASA framework, and best practices for communicating analytical objectives.

Data Project Questions & Considerations

APPASA: Ask Stage

Business Understanding

- Who are the key stakeholders involved in this project?

The key stakeholders involved in this project are:

• **Lily Moreno (Director of Marketing):** As the project sponsor and my manager, Lily Moreno is responsible for developing marketing campaigns and strategic initiatives to promote the Cyclistic bike-share program. She has tasked the marketing analytics team with understanding how annual members and casual riders use Cyclistic bikes differently to support the development of targeted membership conversion strategies.

• **Cyclistic Marketing Analytics Team:** As a junior data analyst, I am part of the marketing analytics team responsible for collecting, analyzing, and communicating data-driven insights that support Cyclistic's marketing objectives. The team's findings will guide the development of evidence-based marketing recommendations.

• **Cyclistic Executive Team:** The executive team will review the analytical findings, visualizations, and recommendations before approving any proposed marketing strategy. They require clear, accurate, and well-supported insights to make informed business decisions regarding future membership growth initiatives.

These stakeholders play a critical role throughout the project, from defining the business objectives to evaluating the final recommendations aimed at increasing the number of annual memberships.

- What business problem is Cyclistic trying to solve?

Cyclistic is trying to address the challenge of increasing the number of annual memberships to support the company's long-term growth and profitability. While the company's flexible pricing structure, including single-ride passes, full-day passes, and annual memberships, has successfully attracted a broad customer base, Cyclistic's finance analysts have concluded that annual members are significantly more profitable than casual riders.

Rather than focusing its marketing efforts on acquiring entirely new customers, Cyclistic aims to convert existing casual riders into annual members. Since casual riders are already familiar with the bike-share program and actively use its services, they represent a valuable opportunity for sustainable membership growth.

To achieve this objective, Cyclistic must first understand how annual members and casual riders differ in their riding behavior. These insights will enable the marketing team to develop targeted, data-driven marketing strategies that encourage casual riders to purchase annual memberships while supporting informed decision-making by the executive team.

- What are the primary business objectives of this project?

The primary business objective of this project is to support Cyclistic's long-term growth by increasing the number of annual memberships. To achieve this, the project aims to identify and understand the behavioral differences between annual members and casual riders through the analysis of Cyclistic's historical bike trip data.

By uncovering meaningful patterns in rider behavior, the project seeks to generate actionable insights that will help the marketing team develop targeted strategies for converting casual riders into annual members. These findings will also provide the executive team with reliable, data-driven evidence to evaluate and approve future marketing initiatives.

Ultimately, the project aims to transform historical trip data into strategic business insights that support informed decision-making, improve marketing effectiveness, and contribute to sustainable membership growth.

- What business task has been assigned to me?

As a junior data analyst on the Cyclistic marketing analytics team, I have been assigned to investigate how annual members and casual riders use Cyclistic bikes differently. This analysis focuses on the first of the three business questions identified by the marketing team and serves as the foundation for developing future membership conversion strategies.

The business question assigned to me is:

1. How do annual members and casual riders use Cyclistic bikes differently?

My responsibility is to analyze Cyclistic's historical bike trip data, identify meaningful behavioral differences between annual members and casual riders, and communicate these findings through clear, data-driven insights and professional visualizations. Based on the analysis, I will provide actionable recommendations that support the marketing team's objective of converting casual riders into annual members and assist the executive team in making informed business decisions.

- What business question(s) should this project answer?

This project is guided by three key business questions identified by the Cyclistic marketing team:

1. **How do annual members and casual riders use Cyclistic bikes differently?**

2. **Why would casual riders purchase Cyclistic annual memberships?**

3. **How can Cyclistic use digital media to encourage casual riders to become annual members?**

As the assigned data analyst, my primary responsibility is to answer the first question by analyzing Cyclistic's historical bike trip data to identify behavioral differences between annual members and casual riders. The insights generated from this analysis will serve as the foundation for addressing the remaining business questions and developing targeted marketing strategies aimed at increasing annual memberships.

- Why is this project important to Cyclistic?

This project is important to Cyclistic because increasing the number of annual memberships is a key driver of the company's long-term growth and profitability. Although Cyclistic's flexible pricing model has successfully attracted both casual riders and annual members, the company's finance analysts have determined that annual members generate greater long-term value than casual riders.

By understanding how annual members and casual riders differ in their riding behavior, Cyclistic can develop more effective and targeted marketing strategies that encourage casual riders to purchase annual memberships. This approach allows the company to focus on converting existing customers who are already familiar with the service, rather than relying solely on acquiring new customers.

The insights generated through this project will support data-driven decision-making, strengthen future marketing initiatives, and contribute to sustainable business growth by increasing customer retention and membership adoption.

- How can the insights generated from this project support business decisions?

The insights generated from this project will enable Cyclistic to make informed, data-driven business decisions by providing a deeper understanding of how annual members and casual riders differ in their riding behavior. Identifying these behavioral patterns will help the marketing team design targeted campaigns and promotional strategies that address the specific needs and preferences of casual riders.

The findings will also support the executive team in evaluating and approving marketing initiatives by providing clear evidence to justify strategic decisions. By understanding when, where, and how different customer groups use Cyclistic bikes, the company can allocate marketing resources more effectively, improve membership conversion efforts, and maximize the long-term profitability of its customer base.

Ultimately, these insights will help Cyclistic develop marketing strategies that encourage more casual riders to become annual members, supporting sustainable business growth and increasing the overall value generated from its existing customers.

- What deliverables will be produced throughout this project?

Throughout this project, I will produce a comprehensive set of technical and business-focused deliverables that document the complete analytics workflow and communicate the project's findings effectively. These deliverables include:

• **A project proposal** outlining the project's objectives, scope, deliverables, estimated timeline, and planned analytical workflow.

• **APPASA strategy documents** for each stage of the project, documenting key decisions, reflections, and planning throughout the analytics workflow.

• **A collection of SQL scripts** covering data ingestion, preprocessing, feature engineering, data validation, and analytical querying.

• **A DuckDB-powered Jupyter Notebook** documenting the complete APPASA methodology, analytical • workflow, SQL queries, visualizations, and business insights.

• **An interactive Power BI report** presenting rider behavior, ride location analysis, and strategic business recommendations.

• **A Power BI report PDF** for offline viewing and presentation purposes.

• **A report walkthrough video** demonstrating the functionality and insights of the Power BI report.

• **An executive summary** highlighting the project's objectives, methodology, key findings, and recommendations.

• **A GitHub repository** containing the project documentation, source code, supporting assets, and final deliverables.

These deliverables collectively demonstrate the complete end-to-end analytics process, from business understanding and project planning to data preparation, analysis, visualization, and strategic decision-making.

Stakeholder Alignment

- Who is the intended audience for this project?

The intended audience for this project includes the Cyclistic marketing analytics team, Lily Moreno, the Director of Marketing, and the Cyclistic executive team. Each audience has a distinct role in the decision-making process and will use the project's findings in different ways.

The marketing analytics team will use the analysis to better understand rider behavior and develop evidence-based marketing strategies. Lily Moreno will use the insights to design initiatives aimed at converting casual riders into annual members and to communicate the value of these strategies to senior leadership. The executive team will review the analytical findings, visualizations, and business recommendations to evaluate and approve future marketing campaigns.

Because the audience consists of both technical and non-technical stakeholders, the findings must be communicated through clear explanations, professional data visualizations, and actionable business recommendations that support informed decision-making.

- What are the expectations of each stakeholder?

Each stakeholder has specific expectations based on their role in the project and how they will use the analytical findings.

• **Lily Moreno (Director of Marketing):** Expects a comprehensive analysis of rider behavior that explains how annual members and casual riders differ. She expects actionable, data-driven recommendations that can be translated into effective marketing strategies for increasing annual memberships.

• **Cyclistic Marketing Analytics Team:** Expects accurate, reliable, and well-documented analyses that provide meaningful insights into customer behavior. The team relies on these findings to support strategic planning, campaign development, and future marketing initiatives.

• **Cyclistic Executive Team:** Expects clear, concise, and evidence-based findings supported by professional data visualizations. Before approving any proposed marketing program, the executive team requires confidence that the recommendations are backed by robust analysis and aligned with Cyclistic's long-term business objectives.

By meeting these expectations, the project will provide stakeholders with the information needed to make informed decisions that support membership growth and improve the effectiveness of Cyclistic's marketing strategy.

- What decisions could stakeholders make based on this analysis?

The findings from this project will enable stakeholders to make informed business decisions aimed at increasing the number of annual memberships and improving the effectiveness of Cyclistic's marketing strategy. By understanding how annual members and casual riders differ in their riding behavior, stakeholders can identify opportunities to better engage casual riders and encourage long-term membership adoption.

Based on the analysis, stakeholders may decide to develop targeted marketing campaigns for specific rider segments, introduce promotional offers or membership incentives, optimize digital marketing strategies, and allocate marketing resources toward locations, seasons, or time periods with the greatest conversion potential. The insights may also influence future pricing strategies, customer engagement initiatives, and partnership opportunities designed to attract and retain annual members.

Ultimately, these decisions will help Cyclistic strengthen customer retention, maximize the value of its existing customer base, and support sustainable business growth through data-driven marketing initiatives.

Project Planning

- What analytical approach will I follow to complete this project?

To complete this project, I will follow the APPASA data analytics methodology, which consists of six sequential stages: Ask, Prepare, Process, Analyze, Share, and Act. This structured approach provides a systematic framework for understanding the business problem, preparing and processing the data, performing meaningful analyses, communicating insights, and developing actionable business recommendations.

Throughout the project, I will use SQL to perform data ingestion, preprocessing, feature engineering, data validation, and analytical querying. A DuckDB-powered Jupyter Notebook will document the complete analytical workflow, reproduce the SQL analyses, and present supporting visualizations and interpretations. Finally, I will develop an interactive Power BI report to communicate the project's findings through business-focused dashboards, data storytelling, and strategic recommendations.

By following this structured methodology, I aim to ensure that every stage of the project is well documented, reproducible, and aligned with Cyclistic's business objectives while delivering reliable, data-driven insights that support informed decision-making.

- What tools and technologies are expected to be used throughout this project?

This project will leverage a combination of database, analytics, visualization, and documentation tools to support the complete end-to-end data analytics workflow. Each tool has been selected based on its strengths and role within the APPASA methodology.

The primary tools and technologies expected to be used throughout this project include:

• **MySQL:** For data ingestion, preprocessing, transformation, feature engineering, data validation, and analytical querying.

• **DuckDB:** For executing SQL queries within a Jupyter Notebook and documenting the complete analytical workflow.

• **Jupyter Notebook:** For documenting the APPASA methodology, SQL analyses, visualizations, interpretations, and business insights.

• **Python:** For creating analytical visualizations and supporting the notebook environment where required.

• **Power BI:** For developing an interactive business intelligence report that communicates key findings and strategic recommendations.

• **GitHub:** For hosting the project repository, documentation, SQL scripts, notebooks, Power BI assets, supporting project files, and managing project versions.

Together, these tools provide a robust and reproducible analytics workflow, enabling efficient data processing, comprehensive analysis, effective visualization, and clear communication of business insights.

- What defines the successful completion of this project?

The successful completion of this project will be defined by its ability to answer the assigned business question and provide actionable, data-driven recommendations that support Cyclistic's objective of increasing annual memberships. Success will be measured not only by the quality of the analysis, but also by the clarity, reliability, and practical value of the insights delivered to stakeholders.

Specifically, the project will be considered successful if it:

• Clearly identifies and explains the behavioral differences between annual members and casual riders.

• Produces accurate, well-documented, and reproducible analyses supported by SQL queries and appropriate data validation.

• Communicates findings effectively through professional visualizations, an interactive Power BI report, and an executive summary.

• Provides strategic business recommendations that can help the marketing team design initiatives to convert casual riders into annual members.

• Demonstrates a complete end-to-end analytics workflow by following the APPASA methodology, from business understanding through actionable recommendations.

By achieving these objectives, the project will deliver meaningful business value while demonstrating a structured, professional, and reproducible approach to solving a real-world data analytics problem.

Reflection

- What assumptions or expectations do I currently have before beginning the project?

Before beginning the analysis, I expect that annual members and casual riders will exhibit distinct riding patterns and usage behaviors. Based on the project scenario, I anticipate that annual members are more likely to use Cyclistic bikes for regular commuting and routine travel, whereas casual riders are more likely to use the service for leisure, recreation, or occasional trips.

I also expect to observe differences in ride duration, ride frequency, preferred ride times, seasonal usage, bike type preferences, and popular start and end stations between the two rider groups. These behavioral differences may provide valuable opportunities for developing targeted marketing strategies that encourage casual riders to purchase annual memberships.

While these expectations provide an initial direction for the analysis, they will be validated or refuted through a comprehensive examination of Cyclistic's historical bike trip data. Any conclusions drawn from this project will be based solely on evidence obtained through data analysis rather than prior assumptions.

Data Project Questions & Considerations

APPASA: Prepare Stage

Data Source

- What dataset(s) will be used for this project?

This project uses **Cyclistic's historical bike trip data**, represented by the **twelve monthly Divvy trip datasets** covering **January 2024 through December 2024**. Each monthly dataset is distributed as a compressed ZIP archive containing a CSV file of individual bike trip records. Together, these twelve monthly datasets represent a complete year of bike-share activity and collectively form the primary dataset used throughout this analysis.

Each trip record represents a single bike trip and contains information including the ride identifier, bicycle type, trip start and end timestamps, station names and identifiers, geographic coordinates, and rider classification (annual member or casual rider). Collectively, these variables provide the information required to examine rider behavior, identify usage patterns, compare annual members with casual riders, and answer the project's primary business question.

- Where is the data located, and how will it be stored?

The dataset is publicly available through the **official Divvy Trip Data repository**, where historical Divvy bike trip data is distributed as monthly ZIP archives containing CSV files. For this project, I downloaded the twelve monthly datasets covering **January 2024 through December 2024** directly from the official repository.

After downloading, the datasets will be organized within a structured local project directory, preserving both the original ZIP archives and the extracted CSV files. The extracted CSV files will then be imported into a **MySQL** database for data ingestion, preprocessing, validation, feature engineering, and analytical querying. Throughout the project, SQL scripts, project documentation, Jupyter notebooks, and supporting assets will be maintained within the project repository, while the processed data will remain stored in the MySQL database to support a reproducible, organized, and efficient analytical workflow.

Data Governance

- Who published the dataset, and what is its source?

The dataset used in this project is publicly available through the **official Divvy Trip Data repository**, which serves as the official source for historical Divvy bike trip data. The files are hosted on an Amazon S3 storage bucket and distributed as monthly ZIP archives containing trip-level CSV datasets.

The **Google Data Analytics Capstone** states that the dataset is made available by **Motivate International Inc.** under the applicable usage license. According to the current **Divvy Data License Agreement**, however, the **City of Chicago** owns the dataset, while **Lyft Bikes and Scooters, LLC.**, the current operator of the Divvy bike-share system, makes the data publicly available through the official Divvy data portal.

For this project, I used the twelve monthly datasets covering **January 2024 through December 2024**, which provide comprehensive records of bike-share trips over a one-year period.

The Google Data Analytics Capstone references these publicly available Divvy datasets because the fictional company **Cyclistic** is modeled using real-world Divvy bike-share data. As a result, the Divvy trip datasets provide an appropriate and representative data source for answering the project's business questions.

- How are licensing, privacy, security, and accessibility addressed for this dataset?

The dataset used in this project is publicly available through the official Divvy Trip Data repository and is distributed under the **Divvy Data License Agreement**. The license grants users a non-exclusive, royalty-free, perpetual license to access, reproduce, analyze, modify, and use the data for lawful purposes, including educational and analytical projects. However, the license prohibits redistributing the dataset as a standalone resource, attempting to identify individual riders, circumventing data access restrictions, or implying any affiliation with or endorsement by Divvy, Lyft Bikes and Scooters, LLC, or the City of Chicago.

The dataset has been anonymized and does not contain any personally identifiable information (PII). In accordance with the Divvy Data License Agreement, users must not attempt to correlate the trip data with names, addresses, payment information, or any other information that could identify individual riders. This ensures that rider privacy is protected while still enabling meaningful behavioral analysis.

To maintain data security throughout this project, the downloaded datasets will be stored within a structured local project directory and imported into a MySQL database for analysis. The original source files will be preserved separately from the processed data to maintain data integrity, support reproducibility, and provide an auditable record of the raw data. Throughout the project, only aggregated analytical results, visualizations, and business insights will be presented, ensuring that the analysis remains ethical, privacy-conscious, and compliant with the dataset's licensing terms.

Because the dataset is publicly available through the official Divvy Trip Data repository, it is readily accessible for educational, research, and analytical purposes. This accessibility enables the dataset to be downloaded, reproduced, and analyzed while supporting transparent, reproducible, and well-documented analytics workflows.

Dataset Understanding

- Why is this dataset appropriate for answering the business question?

This dataset is appropriate for answering the project's business question because it contains detailed historical records of individual bike trips made by both **annual members** and **casual riders**, the two customer groups central to this analysis. Each trip record includes attributes such as rider classification, bicycle type, trip start and end timestamps, station information, and geographic coordinates, providing the information necessary to compare how the two rider groups use Cyclistic bikes.

By covering a complete twelve-month period from **January 2024 through December 2024**, the dataset captures seasonal variations, weekday and weekend travel patterns, ride durations, station usage, and other behavioral trends. This comprehensive temporal coverage enables meaningful comparisons between annual members and casual riders while minimizing the influence of short-term fluctuations or seasonal bias.

The dataset represents real-world operational bike-share data collected from the Divvy system, which serves as the basis for the fictional Cyclistic case study in the Google Data Analytics Capstone. As a result, it provides a realistic and representative foundation for identifying rider behavior patterns, generating actionable business insights, and developing data-driven marketing strategies aimed at converting casual riders into annual members.

- How is the dataset organized?

The dataset is organized as **twelve separate monthly CSV files**, each representing all recorded Divvy bike trips for a specific month between **January 2024 and December 2024**. Together, these monthly datasets form a complete annual dataset that captures one full year of bike-share activity.

Each CSV file follows a **consistent tabular structure**, where **each row represents a single bike trip** and **each column represents a specific attribute** describing that trip. The dataset contains 13 variables, including the ride identifier, bicycle type, trip start and end timestamps, station names and identifiers, geographic coordinates, and rider classification (annual member or casual rider).

Because every monthly dataset shares the same schema, the files can be seamlessly combined into a single consolidated annual dataset without requiring structural modifications. This consistent organization supports efficient data ingestion, preprocessing, validation, feature engineering, analytical querying, and visualization throughout the project's analytical workflow.

- What files comprise the dataset?

The dataset comprises **twelve monthly Divvy trip datasets**, each distributed as a compressed ZIP archive containing a CSV file of individual bike trip records. Collectively, these files represent a complete year of historical bike-share activity spanning **January 2024 through December 2024**.

The dataset includes the following monthly CSV files:

• 202401-divvy-tripdata.csv

• 202402-divvy-tripdata.csv

• 202403-divvy-tripdata.csv

• 202404-divvy-tripdata.csv

• 202405-divvy-tripdata.csv

• 202406-divvy-tripdata.csv

• 202407-divvy-tripdata.csv

• 202408-divvy-tripdata.csv

• 202409-divvy-tripdata.csv

• 202410-divvy-tripdata.csv

• 202411-divvy-tripdata.csv

• 202412-divvy-tripdata.csv

Each file contains all recorded bike trips for its respective month and follows an identical schema consisting of **13 variables**. This consistent structure enables the monthly datasets to be efficiently combined into a single annual dataset for preprocessing, analysis, visualization, and reporting while maintaining schema consistency and data integrity throughout the analytical workflow.

- What variables are available for analysis?

The dataset contains **13 variables** that collectively describe the characteristics of each recorded bike trip. These variables provide the information necessary to analyze rider behavior, usage patterns, and operational characteristics of the Cyclistic bike-share system.

The available variables can be grouped into the following categories:

• **Trip Identification**

- ride_id – Unique identifier assigned to each bike trip.

• **Bicycle Information**

- rideable_type – Type of bicycle used during the trip.

• **Temporal Information**

- started_at – Date and time when the trip began.

ended_at – Date and time when the trip ended.

• **Station Information**

- start_station_name – Name of the origin station.

- start_station_id – Unique identifier of the origin station.

- end_station_name – Name of the destination station.

- end_station_id – Unique identifier of the destination station.

• **Geographic Information**

- start_lat – Latitude of the trip origin.

- start_lng – Longitude of the trip origin.

- end_lat – Latitude of the trip destination.

- end_lng – Longitude of the trip destination.

• **Rider Classification**

- member_casual – Indicates whether the rider is an annual member or a casual rider.

Collectively, these variables provide the information required to examine temporal, spatial, and behavioral patterns, compare annual members with casual riders, and answer the project's primary business question.

- What is the data dictionary for this project?

The data dictionary is a structured reference that documents every variable contained in the dataset, including its name, data type, description, and purpose. It serves as a central source of metadata, ensuring a consistent understanding of the dataset throughout the project's lifecycle while supporting accurate interpretation, preprocessing, analysis, and reporting.

For this project, the data dictionary documents the 13 variables contained within the Divvy trip dataset, including ride identifiers, bicycle types, trip timestamps, station information, geographic coordinates, and rider classifications. It provides a standardized reference for understanding each variable's data type, meaning, and intended use, helping maintain consistency across SQL queries, feature engineering, visualizations, and business reporting.

The table below presents the complete data dictionary for the variables used throughout this project.

Variable	Data Type	Description
ride_id	VARCHAR	Unique identifier assigned to each bike trip.
rideable_type	VARCHAR	Type of bicycle used for the trip (e.g., classic bike or electric bike).
started_at	TIMESTAMP	Date and time when the trip began.
ended_at	TIMESTAMP	Date and time when the trip ended.
start_station_name	VARCHAR	Name of the station where the trip originated.
start_station_id	VARCHAR	Unique identifier of the origin station.
end_station_name	VARCHAR	Name of the station where the trip ended.
end_station_id	VARCHAR	Unique identifier of the destination station.
start_lat	DOUBLE	Latitude coordinate of the trip origin.
start_lng	DOUBLE	Longitude coordinate of the trip origin.
end_lat	DOUBLE	Latitude coordinate of the trip destination.
end_lng	DOUBLE	Longitude coordinate of the trip destination.
member_casual	VARCHAR	Rider classification indicating whether the rider is an annual member or a casual rider.

Data Credibility

- Does the dataset satisfy the ROCCC framework?

The **ROCCC framework** is used to evaluate the credibility and suitability of a dataset by assessing whether it is **Reliable, Original, Comprehensive, Current, and Cited**. Based on this evaluation, the Divvy trip dataset satisfies all five components of the ROCCC framework and provides a credible foundation for answering the project's business question.

Reliable:
The dataset consists of real-world operational records generated by the Divvy bike-share system and is publicly distributed through the official Divvy Trip Data repository. Its official origin, documented licensing, and widespread use in the Google Data Analytics Capstone make it a reliable source for behavioral analysis.

Original:
The data is obtained directly from the official Divvy Trip Data repository rather than a third-party aggregator or secondary source. This ensures that the dataset represents the original operational records collected by the bike-share system.

Comprehensive:
The dataset contains detailed trip-level information, including ride identifiers, bicycle types, trip timestamps, station information, geographic coordinates, and rider classifications. By covering a complete twelve-month period from **January 2024 through December 2024**, it provides sufficient temporal coverage to analyze rider behavior across different seasons and usage patterns.

Current:
The project uses the most recent complete calendar year available at the time the analysis was conducted, covering **January 2024 through December 2024**. This provides a recent and representative view of rider behavior while capturing seasonal variations across an entire year.

Cited:
The dataset is publicly available through the official Divvy Trip Data repository and is referenced by the Google Data Analytics Capstone as the data source for the Cyclistic case study. Its source, ownership, publisher, and licensing information are clearly documented, allowing the dataset to be properly cited and independently verified.

Based on this evaluation, the dataset satisfies all five components of the ROCCC framework and is well suited for conducting reliable, reproducible, and data-driven analysis.

- Are there any potential sources of bias or limitations within the dataset?

Although the dataset is comprehensive and well suited for this analysis, several limitations should be considered when interpreting the results.

The dataset has been **anonymized** and does not contain personally identifiable information (PII) or demographic attributes such as age, gender, income, occupation, or place of residence. As a result, the analysis can identify behavioral differences between annual members and casual riders but cannot determine how demographic or socioeconomic factors influence those differences.

The dataset records **observed bike trips** rather than customer motivations or preferences. Consequently, it can reveal **how** annual members and casual riders behave differently but cannot directly explain **why** those behavioral differences exist or what factors motivate casual riders to purchase an annual membership.

Additionally, the analysis is based on historical trip records covering a single twelve-month period from **January 2024 through December 2024**. While this provides complete seasonal coverage for one calendar year, changes in rider behavior, pricing policies, infrastructure, or external conditions outside this period may influence future travel patterns.

Finally, the dataset describes completed bike trips and their associated characteristics. It does not include complementary business information such as marketing campaign exposure, customer feedback, membership purchase history, or financial metrics, limiting the ability to directly evaluate the effectiveness of specific marketing strategies or the profitability of individual rider segments.

These limitations do not prevent meaningful analysis but should be considered when interpreting the findings and developing business recommendations.

- Are there any ethical considerations associated with using this dataset?

Yes. Although the dataset is publicly available and anonymized, several ethical considerations should be observed throughout the project.

The dataset does not contain personally identifiable information (PII), and rider privacy must be respected at all times. In accordance with the **Divvy Data License Agreement**, no attempt should be made to correlate the trip records with names, addresses, payment information, or any other external data that could be used to identify individual riders.

The analysis should focus on **aggregated trends and behavioral patterns** rather than individual trips or riders. All findings, visualizations, and recommendations should be presented at an aggregate level to ensure that no individual can be identified, directly or indirectly.

It is also important to interpret the results objectively and avoid drawing conclusions that are not supported by the available data. Since the dataset records only observed trip behavior, the analysis should not make unsupported assumptions about riders' motivations, preferences, demographics, or personal circumstances.

Finally, the dataset should be used in accordance with the **Divvy Data License Agreement**, respecting its licensing conditions and usage restrictions while ensuring that all analyses, reports, and project deliverables are conducted responsibly and ethically.

Data Suitability & Readiness

- How does this dataset support the project's business objective?

The primary business objective of this project is to support Cyclistic's goal of increasing the number of annual memberships by understanding how annual members and casual riders use the bike-share system differently. The dataset directly supports this objective by providing detailed historical records of bike trips taken by both rider groups over a complete twelve-month period.

Because each trip record includes rider classification, bicycle type, trip timing, station information, and geographic coordinates, the dataset enables meaningful comparisons of riding behavior, travel patterns, trip durations, bicycle preferences, and station usage between annual members and casual riders. These behavioral insights can help identify opportunities to encourage casual riders to transition to annual memberships.

By transforming raw trip data into actionable business insights, the dataset provides the evidence needed to develop targeted marketing strategies, support data-driven decision-making, and formulate recommendations that align with Cyclistic's long-term objective of increasing annual membership adoption.

- How was the data's integrity initially verified?

Before beginning data processing, the dataset's integrity was initially verified by confirming that all required files were successfully downloaded from the official Divvy Trip Data repository and corresponded to the intended analysis period of **January 2024 through December 2024**. Each monthly dataset was extracted from its ZIP archive and inspected to ensure that the files were complete, accessible, and free from corruption.

An initial review was then conducted to verify that all twelve CSV files shared a consistent schema, including identical variable names, data types, and column ordering. The datasets were also checked to confirm that each file contained trip-level records with the expected structure and could be successfully imported into the project's analytical environment without structural issues.

These initial verification steps established that the dataset was complete, structurally consistent, and ready to proceed to the data processing stage, where detailed data quality assessments and cleaning operations would be performed.

- What potential data quality issues should be expected before processing begins?

Before beginning the data processing stage, several potential data quality issues can reasonably be anticipated due to the size and operational nature of the dataset. These expectations help guide the data validation and cleaning process that follows.

Potential issues include:

• **Missing values:** Certain fields, such as station names, station identifiers, or geographic coordinates, may contain missing or incomplete information.

• **Duplicate records:** Duplicate trip records may exist because of data collection or processing anomalies and should be identified during validation.

• **Invalid or inconsistent timestamps:** Trip start and end timestamps should be examined for formatting inconsistencies or unrealistic ride durations.

• **Data type inconsistencies:** Variables should be verified to ensure they conform to their expected data types and formats.

• **Schema consistency:** Although the monthly datasets are expected to share an identical schema, the column structure and naming conventions should be verified before combining the files.

• **Incomplete or erroneous records:** Some trip records may contain incomplete, inconsistent, or invalid values that could affect the accuracy of subsequent analyses.

These potential issues represent expected data quality considerations rather than confirmed findings. A comprehensive assessment of data quality will be performed during the **Process** stage before any analytical activities begin.

- What assumptions do I currently have about the dataset before beginning the processing stage?

Before beginning the data processing stage, I assume that the dataset is generally well structured, complete, and suitable for analysis because it is obtained directly from the official Divvy Trip Data repository and is used as the primary dataset for the Google Data Analytics Capstone.

I also assume that the twelve monthly datasets share a consistent schema, allowing them to be combined into a single annual dataset without requiring significant structural modifications. Although I expect the dataset to be of high quality, I anticipate encountering common real-world data quality issues such as missing values, duplicate records, inconsistent timestamps, or incomplete trip information that will require validation and cleaning during the Process stage.

Finally, I assume that the available variables contain sufficient information to identify meaningful behavioral differences between annual members and casual riders. However, I recognize that these assumptions must be verified through systematic data exploration, validation, and preprocessing before any analysis or business conclusions are drawn.

Data Project Questions & Considerations

APPASA: Process Stage

Data Processing Environment

- What tools were selected for data processing, and why?

The primary tool selected for data processing was **MySQL**, which served as the project's data processing engine. MySQL was chosen because it provides a robust relational database management system capable of efficiently handling millions of records while supporting SQL-based data ingestion, validation, cleaning, transformation, and feature engineering. Its support for Common Table Expressions (CTEs), window functions such as `ROW_NUMBER()`, indexing, and built-in date and string functions enabled the entire data preparation workflow to be performed within the database, creating a clean and reproducible ELT pipeline.

MySQL Workbench was used as the development environment for writing, executing, and documenting SQL scripts, inspecting intermediate results, and validating data quality throughout the processing workflow.

The raw datasets were initially stored as **CSV files**, preserving the original source data exactly as downloaded before being imported into MySQL. Maintaining the raw files separately from the processed database tables ensured data integrity, reproducibility, and an auditable processing pipeline. This separation also allowed the raw staging table to remain unchanged while a cleaned analytical table was created for downstream analysis.

Overall, this toolset supported an efficient, scalable, and well-documented data processing workflow that prepared the dataset for the subsequent Analyze stage.

- Why was MySQL chosen as the primary data processing environment?

MySQL was chosen as the primary data processing environment because it is specifically designed to efficiently manage and process large structured datasets using SQL. The Cyclistic dataset contained over **5.8 million trip records**, making a relational database management system more suitable than spreadsheet-based tools for scalable data processing.

MySQL enabled the entire processing workflow—including data ingestion, validation, cleaning, feature engineering, and data quality verification—to be performed within a single environment. Its support for Common Table Expressions (CTEs), window functions (`ROW_NUMBER()`), indexing, and built-in string, date, and conditional functions made it possible to implement a robust and reproducible ELT pipeline while preserving the raw dataset.

Using MySQL also separated the data processing stage from the analysis and visualization stages. Rather than repeatedly performing transformations within the business intelligence tool, a clean, processed analytical table was created once in the database and subsequently used for downstream analysis in Power BI. This approach improves maintainability, reproducibility, and processing efficiency while aligning with industry practices for managing large analytical datasets.

- How does the ELT workflow support this project?

The project follows an **Extract, Load, Transform (ELT)** workflow to preserve the integrity of the original data while creating a clean analytical dataset. First, the twelve monthly Divvy trip datasets were extracted from their source and loaded into a raw staging table within MySQL without modification. All subsequent data validation, cleaning, feature engineering, and quality verification were performed directly within the database, producing a separate processed table for downstream analysis.

This approach ensures that the original data remains unchanged, allowing every transformation to be reproduced, audited, or revised without re-importing the source files. It also separates raw and processed data, making the workflow more maintainable and reducing the risk of accidental data loss or modification. By performing transformations inside MySQL, the project leverages the database engine to efficiently process millions of records before the cleaned dataset is imported into Power BI for analysis and visualization.

Data Validation

- What validation checks were performed before cleaning?

Before applying any cleaning operations, a comprehensive data validation process was performed to assess the integrity, completeness, and consistency of the raw dataset. The validation phase began by verifying the dataset structure, including the total number of imported records and the table schema, to confirm that the data had been imported successfully.

The dataset was then examined for duplicate records by identifying duplicate `ride_id` values and manually investigating representative duplicate patterns to determine an appropriate deduplication strategy. Missing values were assessed by checking for both `NULL` values and blank strings in station names and station IDs. Destination coordinate fields were also validated to identify zero-valued latitude and longitude records representing missing spatial information.

Finally, temporal integrity checks were performed to detect records with invalid timestamps, including trips ending before they started, negative-duration trips, zero-duration trips, and trips exceeding 24 hours. These validation findings provided the evidence required to define the cleaning rules applied during the subsequent data cleaning stage.

- What data quality issues were identified?

The validation process identified several data quality issues that required remediation before analysis. Duplicate `ride_id` values were detected, consisting of both exact duplicate records and duplicate records with slight variations in the `ended_at` timestamp. Missing station information was also identified, with station names and station IDs represented as blank strings rather than standardized `NULL` values.

The destination coordinate fields contained records where both `end_lat` and `end_lng` were recorded as `0`, indicating missing geographic information instead of valid coordinates. Further investigation revealed that the majority of these records also exhibited temporal anomalies and were subsequently excluded during temporal cleaning, while the remaining valid records had their missing coordinates standardized to `NULL`.

Temporal validation also identified trips with invalid timestamps, including rides ending before they started, non-positive ride durations, and rides exceeding the project's maximum accepted duration of 24 hours. These issues informed the cleaning rules implemented to produce a reliable analytical dataset for downstream analysis.

- Why was it important to validate the raw dataset before cleaning?

Validating the raw dataset before cleaning ensured that all data quality issues were identified and understood before any modifications were made. Rather than applying generic cleaning operations, the validation phase provided evidence for each cleaning decision by quantifying issues such as duplicate records, missing values, invalid spatial data, and temporal anomalies.

This evidence-based approach helped prevent unnecessary or incorrect transformations while ensuring that every cleaning rule addressed a specific, verified data quality issue. It also preserved the integrity of the raw dataset by allowing the original data to remain unchanged throughout the investigation. As a result, the cleaning process became transparent, reproducible, and well-documented, with each transformation justified by the findings of the validation phase.

- How were duplicate records investigated before deciding on a cleaning rule?

Duplicate records were first identified by detecting repeated `ride_id` values within the raw dataset. Rather than immediately removing these records, representative duplicate groups were manually inspected to understand the nature of the duplication. This investigation revealed two distinct patterns: exact duplicate records, where all attribute values were identical, and timestamp-variant duplicates, where records shared the same `ride_id` but differed slightly in the `ended_at` timestamp.

Based on these findings, a deterministic deduplication strategy was established. The record with the most recent `ended_at` timestamp was retained for each `ride_id`, while all remaining duplicate records were excluded from the processed dataset. This approach ensured that every ride was represented by a single record while applying a transparent and reproducible retention rule supported by the validation findings.

Data Cleaning

- What cleaning rules were applied to the dataset??

The cleaning process applied a series of rule-based transformations to improve data quality while preserving the integrity of the original dataset. Duplicate records were resolved by retaining a single record for each `ride_id` based on the most recent `ended_at` timestamp. Blank station names and station IDs were standardized to `NULL` to ensure consistent representation of missing values.

Destination coordinates with values of `0` for both `end_lat` and `end_lng`, representing missing geographic information, were converted to `NULL`. Records with invalid temporal values were excluded from the processed dataset, including trips ending before they started and trips with durations exceeding the project's maximum accepted threshold of 24 hours.

These cleaning rules were applied during the transformation from the raw staging table to the processed analytical table, ensuring that the resulting dataset was consistent, reliable, and suitable for downstream analysis.

- Why were these cleaning rules selected?

The cleaning rules were selected based on the findings of the data validation phase rather than being applied as generic preprocessing steps. Each rule addressed a specific data quality issue identified during validation, ensuring that every transformation was evidence-based and justified. Duplicate records were resolved using a deterministic retention strategy, missing values were standardized to ensure consistent representation across the dataset, and invalid temporal records were excluded because they did not represent valid trips.

The objective was to improve the quality, consistency, and reliability of the dataset while preserving as much valid information as possible. By applying only validated and well-documented cleaning rules, the project minimized unnecessary data loss and produced a processed analytical dataset suitable for downstream analysis and visualization.

- Which records were removed rather than corrected?

Only records that could not be reliably corrected while preserving data integrity were excluded from the processed dataset. This included duplicate records retained under the project's deduplication strategy, where only one record was preserved for each `ride_id`, as well as records with invalid temporal values that did not represent valid trips. Specifically, trips ending before they started and trips exceeding the maximum accepted duration of 24 hours were removed from the processed dataset.

In contrast, issues that could be standardized without losing valid information were corrected rather than removed. For example, blank station names and station IDs were converted to `NULL`, and zero-valued destination coordinates representing missing geographic information were also standardized to `NULL`. This approach minimized unnecessary data loss while ensuring the analytical dataset remained accurate, consistent, and reliable.

- How was the original raw dataset preserved?

The original raw dataset was preserved by following an Extract, Load, Transform (ELT) workflow. The twelve monthly Divvy trip datasets were first imported into a dedicated raw staging table within MySQL without applying any modifications. All subsequent validation, cleaning, feature engineering, and data quality verification were performed using SQL transformations that populated a separate processed analytical table, leaving the raw table unchanged throughout the project.

Maintaining separate raw and processed tables ensured that the original source data remained available for auditing, revalidation, or future processing if required. This separation also improved reproducibility by allowing the entire transformation pipeline to be rerun from the unchanged raw dataset whenever modifications to the cleaning or feature engineering logic were needed.

- Why was a separate processed table created?

A separate processed table was created to provide a clean, analysis-ready dataset while preserving the integrity of the original raw data. Rather than modifying the source records directly, all cleaning operations, feature engineering, and data quality improvements were applied during the transformation from the raw staging table to the processed analytical table.

This separation ensured that the processed analytical table served as the data source for downstream analysis and dashboard development in Power BI while leaving the raw dataset unchanged. It also improved maintainability and reproducibility by allowing the processing pipeline to be rerun whenever cleaning rules or engineered features required modification, with the raw dataset retained as a permanent reference.

Feature Engineering

- Which analytical features were engineered?

Three analytical features were engineered during the processing stage to support downstream analysis while maintaining a reusable analytical dataset:

• **`ride_length_minutes`** – Calculated as the exact ride duration in minutes from the difference between `started_at` and `ended_at`. This feature supports analyses involving trip duration, rider behavior, and ride length distributions.

• **`ride_date`** – Extracted from the `started_at` timestamp to provide a date-level attribute for temporal analysis and to establish a relationship with the Date dimension table in the Power BI data model.

• **`hour_of_day`** – Extracted from the `started_at` timestamp to represent the ride start hour, enabling analyses of hourly ride demand, peak usage periods, and rider behavior throughout the day.

These features were selected because they represent reusable ride-level attributes that can support a wide range of analytical tasks beyond the project's dashboard, while dashboard-specific classifications and aggregations were intentionally reserved for the Power BI semantic layer.

- Why were these features created?

The engineered features were created to enhance the analytical value of the processed dataset while maintaining a reusable and database-centric data model. Rather than requiring downstream tools to repeatedly derive common analytical attributes, these features were computed once during the processing stage and stored within the processed table.

Each feature represents an intrinsic property of an individual ride rather than a dashboard-specific calculation. Ride duration supports analyses of trip behavior, ride date enables temporal analysis and integration with the Power BI Date dimension, and ride start hour facilitates the analysis of hourly demand and usage patterns. By engineering these reusable attributes within MySQL, the project reduced redundant computations in downstream tools while producing a consistent analytical dataset suitable for a variety of reporting and analytical use cases.

- How do the engineered features support later analysis?

The engineered features provide reusable analytical attributes that simplify downstream analysis and reporting. By deriving these features during the processing stage, subsequent analyses can focus on identifying patterns and generating insights rather than repeatedly performing the same data transformations.

The `ride_length_minutes` feature enables analyses of trip duration, rider behavior, and ride length distributions. The `ride_date` feature supports date-based trend analysis and serves as the relationship key between the processed dataset and the Power BI Date dimension, enabling efficient time-based reporting. The `hour_of_day` feature facilitates analyses of hourly ride demand, peak usage periods, and rider activity throughout the day.

Engineering these features during processing also improves consistency by ensuring that all downstream analyses and visualizations use the same standardized definitions, reducing redundant calculations across analytical workflows.

- Why were these features created during processing rather than analysis?

These features were engineered during the processing stage because they represent reusable ride-level attributes derived directly from the raw data, rather than analytical or visualization-specific calculations. Computing them once during data processing ensures that all downstream analyses, reports, and dashboards use consistent feature definitions without repeatedly performing the same transformations.

Creating these features within MySQL also aligns with the project's ELT architecture by preparing a clean, analysis-ready dataset before it is imported into Power BI. This reduces redundant computations, improves processing efficiency, and allows the Analyze stage to focus on exploring patterns, answering business questions, and generating insights rather than performing fundamental data transformations.

Data Quality Verification

- How was the cleaned dataset verified?

The cleaned dataset was verified through a comprehensive post-cleaning data quality verification process to confirm that all cleaning rules had been successfully applied. Verification queries were executed to reconcile the raw and processed datasets by comparing record counts and quantifying the number of excluded records. Additional checks confirmed that duplicate `ride_id` values had been resolved, missing station fields were consistently represented as `NULL`, zero-valued destination coordinates had been standardized, and all remaining records satisfied the project's temporal integrity requirements.

These verification steps ensured that the processed dataset accurately reflected the intended cleaning rules and was internally consistent before proceeding to feature engineering and downstream analysis.

- How was feature engineering verified?

The engineered features were verified through a dedicated post-engineering verification process to ensure that each feature had been generated accurately and consistently across the processed dataset. Verification queries confirmed that the engineered columns contained no unexpected `NULL` values, that ride durations were correctly calculated from the original timestamps, that ride dates matched the date component of `started_at`, and that ride start hours accurately reflected the hour extracted from the original timestamp.

These validation checks ensured that the engineered features were complete, internally consistent, and correctly derived from the source data before being used for downstream analysis and dashboard development.

- How did you ensure the processed dataset was ready for analysis?

The processed dataset was prepared for analysis through a structured data processing workflow consisting of data validation, cleaning, feature engineering, and post-processing verification. Data validation identified and documented data quality issues before any transformations were applied. Cleaning rules were then implemented to resolve duplicate records, standardize missing values, and exclude invalid temporal records while preserving the original raw dataset. Reusable analytical features were subsequently engineered to enhance the dataset for downstream analysis.

Following these transformations, dedicated verification procedures confirmed that the cleaning rules and engineered features had been applied correctly and consistently. Together, these steps ensured that the processed dataset was accurate, internally consistent, and analysis-ready, providing a reliable foundation for the subsequent Analyze stage and dashboard development.

- How does the verification process improve confidence in the results?

The verification process improves confidence in the results by confirming that the intended data transformations were applied accurately and consistently throughout the processing pipeline. Rather than assuming the cleaning and feature engineering steps executed correctly, dedicated verification queries were used to validate the integrity of the processed dataset after each stage.

By confirming that data quality issues were successfully resolved, engineered features were correctly derived, and the processed dataset satisfied all defined quality requirements, the verification process reduced the risk of undetected errors propagating into subsequent analysis and visualization. This systematic validation increased the reliability, reproducibility, and overall credibility of the project's analytical findings.

Documentation & Reproducibility

- How was the processing workflow documented?

The processing workflow was documented through a structured and fully commented SQL script that records each stage of the data preparation pipeline. The script is organized into logical sections covering project setup, data ingestion, data validation, data cleaning, feature engineering, and data quality verification, with explanatory comments describing the purpose and rationale behind each step.

In addition to the SQL implementation, the processing methodology, validation findings, cleaning decisions, engineered features, and verification procedures were documented within the project's APPASA Process stage documentation. Together, these artifacts provide a transparent, reproducible record of the complete data processing workflow, enabling the project to be reviewed, understood, and replicated by other analysts.

- How does the SQL script support reproducibility?

The SQL script supports reproducibility by providing a structured, deterministic implementation of the entire data processing pipeline. Every stage—from data ingestion and validation to cleaning, feature engineering, and data quality verification—is explicitly defined using SQL statements that can be executed repeatedly to produce the same processed dataset from the same raw data.

Because all transformations are implemented within a single, well-documented script, the complete workflow can be rerun whenever the source data is updated or the processing logic is refined. This eliminates reliance on undocumented manual operations, ensures consistent application of cleaning and transformation rules, and enables other analysts to reproduce the processing workflow and obtain identical results under the same conditions.

- Why is documenting cleaning decisions important?

Documenting cleaning decisions is important because it provides a transparent record of how the raw data was transformed into the final analytical dataset. By clearly recording the rationale behind each cleaning rule, the project enables others to understand what changes were made, why they were necessary, and how they may influence subsequent analysis.

Well-documented cleaning decisions also improve reproducibility, maintainability, and collaboration. They allow the processing workflow to be reviewed, audited, or modified in the future while ensuring that transformations remain consistent and evidence-based. This documentation helps build confidence in the integrity of the processed dataset and supports the credibility of the project's analytical findings.

Data Project Questions & Considerations

APPASA: Analyze Stage

Analysis Environment

- What analytical environment and dataset were used for this stage?

The analytical phase was conducted using **MySQL** as the primary query engine and **MySQL Workbench** as the development environment for writing, executing, and documenting SQL queries. Unlike the Process stage, which focused on data validation, cleaning, transformation, and feature engineering, the Analyze stage operated exclusively on the finalized analytical dataset to perform descriptive statistics, comparative analyses, and business-focused exploration.

The dataset used throughout this stage was the `cyclistic_tripdata_2024_processed` table created during the Process stage. This processed analytical table contains **5,851,692** validated trip records and includes both the original trip attributes and the engineered features developed during data preparation, such as `ride_length_minutes`, `ride_date`, and `hour_of_day`. By using the processed table rather than the raw staging data, all analyses were performed on a clean, standardized, and quality-assured dataset.

The analytical workflow consisted of SQL aggregation functions, conditional logic, window functions, Common Table Expressions (CTEs), and analytical views to examine rider behavior across multiple dimensions, including rider type, ride duration, seasonality, day of the week, hour of the day, bike type, and station usage. This approach produced a reproducible, scalable, and well-documented analysis that directly addressed the project's primary business question while preparing the results for the subsequent Share stage through Power BI visualizations and an executive summary.

- Why was the processed analytical table used instead of the raw dataset?

The processed analytical table was used to ensure that all analyses were performed on a clean, validated, and standardized dataset rather than the raw imported data. During the Process stage, the raw trip records underwent comprehensive data validation, cleaning, transformation, feature engineering, and quality verification to resolve data quality issues and prepare the dataset for analysis. As a result, the processed table contained only reliable records suitable for generating meaningful business insights.

Using a separate processed analytical table also established a clear separation between data preparation and data analysis. The raw staging table remained preserved in its original state, ensuring data integrity and allowing every transformation to be reproduced or audited if necessary. Meanwhile, the Analyze stage could focus exclusively on exploring rider behavior and identifying trends without repeatedly applying cleaning or transformation logic within analytical queries.

This approach also improved the readability, maintainability, and reproducibility of the project. Engineered features such as `ride_length_minutes`, `ride_date`, and `hour_of_day` were already available in the processed dataset, enabling analytical queries to remain concise while ensuring that every result was generated from a consistent and quality-assured data source. This separation aligns with industry best practices for analytical workflows, where data preparation and business analysis are treated as distinct stages within the data lifecycle.

Exploratory Analysis

- How was the analytical baseline established before investigating rider behavior?

Before conducting detailed rider behavior analyses, an analytical baseline was established to develop a comprehensive understanding of the processed dataset and verify that it was suitable for exploration. The analysis began by reviewing the overall size of the dataset, inspecting the table schema, and examining a representative sample of records. These initial checks confirmed the dataset structure, validated that the engineered features were available as expected, and provided familiarity with the analytical variables that would be used throughout the remainder of the project.

Following the dataset overview, descriptive statistics were calculated to summarize the ride duration variable across all processed trips. Key metrics, including the total number of rides, minimum ride duration, maximum ride duration, and average ride duration, provided an initial understanding of the dataset's overall characteristics before segmenting the data by rider type or temporal dimensions.

Establishing this analytical baseline ensured that subsequent analyses were interpreted within the broader context of the entire dataset rather than in isolation. It also provided a consistent reference point against which behavioral differences between annual members and casual riders, temporal usage patterns, bike preferences, and station-level trends could be compared throughout the Analyze stage.

- What descriptive statistics were calculated to understand the overall dataset?

To establish a foundational understanding of the processed analytical dataset, descriptive statistics were calculated for the **ride duration** variable across all processed trips. Specifically, the total number of rides, minimum ride duration, maximum ride duration, and average ride duration were computed using aggregate SQL functions. Together, these metrics provided a concise summary of the dataset's scale, range, and central tendency before conducting more detailed behavioral analyses.

The total ride count quantified the volume of observations available for analysis, while the minimum and maximum ride durations defined the overall range of valid trip lengths after the data cleaning process. The average ride duration served as an initial measure of central tendency, offering a high-level summary of typical trip length across the entire dataset.

Although the average provides a useful overall benchmark, it was not interpreted in isolation. Ride duration data is inherently susceptible to right-skewness due to a relatively small number of exceptionally long trips. Consequently, the average was treated as a descriptive reference point rather than a complete representation of typical rider behavior. This motivated the subsequent ride duration distribution analysis, where rides were grouped into duration intervals to better understand the underlying distribution and assess how representative the mean truly was.

Rider Behavior Analysis

- How do annual members and casual riders differ in their overall riding behavior?

To establish the primary behavioral differences between rider groups, the processed dataset was first segmented by `member_casual` and analyzed using aggregate statistics. Three complementary analyses were performed: total ride volume, each rider type's proportional contribution to overall ride activity, and average ride duration.

The analysis revealed that annual members generated **3,706,732 rides (63.34%)**, while casual riders completed **2,144,960 rides (36.66%)** during the study period. This indicates that annual members account for nearly two-thirds of all trips, suggesting that they are the dominant contributors to Cyclistic's overall ride demand.

Behavioral differences also emerged when comparing average ride duration. Casual riders recorded an average trip length of approximately **20.92 minutes**, compared with **12.19 minutes** for annual members. This suggests that although annual members ride more frequently, casual riders typically spend considerably more time on each trip.

These findings established the first clear distinction between the two rider groups: **annual members exhibit higher ride frequency, whereas casual riders exhibit longer ride durations**. This initial comparison provided the analytical foundation for the subsequent analyses, which further examined how these behavioral differences vary across ride durations, temporal patterns, bike preferences, and station usage.

- How is ride duration distributed, and does the average accurately represent typical rider behavior?

To better understand rider behavior beyond the overall average, ride durations were grouped into meaningful time intervals and analyzed using both absolute ride counts and percentage distributions. Examining the distribution rather than relying solely on the mean provided a more complete understanding of how ride durations were concentrated across the dataset.

The analysis revealed a pronounced **right-skewed distribution**. Approximately **80.80%** of all rides were completed within **20 minutes**, with the largest concentration (**30.23%**) occurring in the **5–10 minute** interval. In contrast, only **2.50%** of rides lasted longer than **60 minutes**, indicating that exceptionally long trips were relatively uncommon.

The distribution was then segmented by rider type to compare behavioral patterns independent of differences in ride volume. Rather than comparing raw counts, percentages were calculated **within each rider type**, allowing the ride duration profiles of annual members and casual riders to be compared on an equal basis despite the larger number of member trips. This analysis showed that annual members were more heavily concentrated in shorter-duration rides, whereas casual riders exhibited a greater proportion of medium- and long-duration trips. For example, **24.22%** of member rides lasted less than five minutes compared with **15.93%** of casual rides, while **5.66%** of casual rides exceeded one hour compared with only **0.66%** of member rides.

These findings demonstrate that the overall average ride duration should be interpreted with caution. Although the average provides a useful high-level summary, it does not accurately represent the typical rider experience because it is influenced by a relatively small number of long-duration trips. The ride duration distribution therefore provides a more representative view of rider behavior while reinforcing the conclusion that annual members primarily use Cyclistic for shorter, routine trips, whereas casual riders are more likely to undertake longer recreational rides.

Temporal Analysis

- How does rider activity vary throughout the year?

To understand how rider activity changes over time, the analysis examined trip patterns across both **monthly** and **seasonal** time periods. Monthly analysis provided a detailed view of fluctuations throughout the calendar year, while seasonal aggregation revealed broader trends associated with changing weather conditions and rider demand.

The results showed a clear seasonal pattern in Cyclistic usage. Ride activity increased steadily from winter into spring, reached its highest levels during the summer months, and gradually declined throughout autumn before reaching its lowest levels during winter. This progression reflects the strong influence of seasonal conditions on bike-share usage, with warmer weather encouraging substantially higher riding activity.

To support seasonal analysis, a reusable analytical view was created to classify each trip into one of four meteorological seasons based on its start month. This enabled ride volume, rider composition, and average ride duration to be analyzed consistently across seasons without repeatedly applying classification logic within individual queries.

Overall, the temporal analysis demonstrates that Cyclistic experiences predictable annual fluctuations in demand, with peak activity occurring during warmer months and reduced usage during colder periods. Establishing these yearly patterns provides important context for understanding when different rider groups are most active and supports the development of seasonally targeted marketing strategies in the subsequent stages of the project.

- How do annual members and casual riders differ across months and seasons?

To compare temporal riding patterns between rider groups, monthly and seasonal analyses were performed separately for annual members and casual riders. These analyses examined ride volume, rider composition, and average ride duration to identify how riding behavior changed throughout the year for each group.

The results showed that both rider types followed the same general seasonal cycle, with activity increasing during the warmer months and declining during colder periods. However, the magnitude of this variation differed considerably between the two groups. Annual members maintained relatively consistent riding activity throughout the year, suggesting that their trips were driven primarily by routine transportation and commuting needs. In contrast, casual riders exhibited much stronger seasonal fluctuations, with ride volumes increasing sharply during spring and summer before declining substantially during autumn and winter.

Differences were also evident in ride duration. While casual riders consistently recorded longer average trips than annual members across all months and seasons, both groups tended to take longer rides during the warmer months. This indicates that favorable weather not only encourages more trips but also supports longer riding sessions, particularly among casual users.

Overall, the analysis suggests that **annual members demonstrate stable, year-round riding behavior**, whereas **casual riders are considerably more influenced by seasonal conditions**. These findings reinforce the distinction between transportation-oriented usage by members and recreational usage by casual riders, providing valuable insight for designing marketing campaigns aimed at converting casual riders into annual members during periods of peak engagement.

- How does rider behavior change throughout the week?

To examine weekly riding patterns, trip activity was analyzed across all seven days of the week using ride volume, rider composition, and average ride duration. Together, these analyses provided insight into how rider behavior varies between weekdays and weekends.

The analysis revealed clear differences in weekly riding behavior between annual members and casual riders. Annual members generated the highest ride volumes on weekdays, particularly during the traditional workweek, indicating that their riding patterns are closely aligned with regular commuting and daily transportation. In contrast, casual riders were more active on weekends, when their share of total rides increased noticeably.

Ride duration analysis further reinforced these behavioral differences. Casual riders consistently recorded longer average trip durations than annual members throughout the week, with the longest rides typically occurring on weekends. Annual members, however, maintained relatively short and stable ride durations across all days, suggesting that their trips were primarily functional and time-efficient.

Overall, the weekly analysis highlights two distinct usage patterns: **annual members exhibit consistent weekday-oriented riding associated with routine travel, whereas casual riders demonstrate stronger weekend engagement and longer recreational trips**. These findings complement the monthly and seasonal analyses by showing that rider behavior varies not only throughout the year but also within a typical week, providing further evidence of fundamentally different usage motivations between the two rider groups.

- How does rider activity vary throughout the day?

To investigate daily riding patterns, trip activity was analyzed by **hour of the day** using ride volume, rider composition, and average ride duration. This analysis identified how demand fluctuates over a typical 24-hour period and revealed differences in riding behavior between annual members and casual riders.

The results showed that rider activity follows a distinct daily cycle, with relatively low ride volumes during the overnight hours, increasing activity throughout the morning, peak demand during daytime and early evening hours, and a gradual decline later in the night. These fluctuations indicate that bike-share usage is strongly influenced by typical daily routines and travel patterns.

Comparing rider types revealed notable differences in when trips occurred. Annual members exhibited pronounced peaks during traditional commuting hours, suggesting that many of their rides were associated with travel to and from work or other routine destinations. Casual riders, on the other hand, displayed a more evenly distributed pattern throughout the day, with higher activity during the afternoon and early evening when recreational and leisure travel is more common.

Average ride duration also varied throughout the day. Casual riders consistently recorded longer trips across most hours, while annual members maintained shorter and more stable ride durations. This pattern further supports the distinction between transportation-oriented usage among members and leisure-oriented usage among casual riders.

Overall, the hourly analysis demonstrates that **rider activity is shaped by daily travel routines**, with annual members exhibiting clear commuter-driven demand and casual riders showing greater flexibility in both ride timing and trip duration. These insights provide a more granular understanding of daily usage patterns and complement the broader temporal trends observed across weeks and seasons.

- What temporal patterns emerge when day of the week and hour of the day are analyzed together?

To obtain a more detailed understanding of rider activity, the day of the week and hour of the day were analyzed together using a two-dimensional temporal analysis. Examining these dimensions simultaneously made it possible to identify recurring demand patterns that would not be apparent when analyzing each variable independently.

The combined analysis revealed distinct temporal clusters in rider behavior. Annual members exhibited concentrated ride activity during weekday morning and evening commuting periods, reflecting consistent work-related travel patterns. In contrast, casual riders showed greater activity during weekend afternoons and early evenings, when recreational riding was most prevalent. These contrasting patterns indicate that the timing of trips is strongly influenced by the underlying purpose of travel.

The analysis also highlighted periods of relatively low demand, particularly during overnight hours across all days of the week. This consistent decline in activity reinforces the cyclical nature of bike-share usage and demonstrates how demand fluctuates according to both daily routines and weekly schedules.

Overall, the combined day-and-hour analysis provides a more comprehensive view of rider behavior than either dimension alone. By revealing **when** different rider groups are most active, it confirms that **annual members predominantly follow weekday commuting schedules**, whereas **casual riders concentrate their activity during weekends and leisure hours**. These insights offer valuable guidance for operational planning, resource allocation, and the timing of marketing initiatives aimed at engaging casual riders during periods of peak recreational demand.

Usage Pattern Analysis

- How do bike type preferences differ between annual members and casual riders?

To understand differences in vehicle preferences, ride activity was analyzed across the available bike types for both annual members and casual riders. The analysis examined total ride volume, rider composition, and average ride duration for each bike type, providing insight into how different bicycles are used by each rider group.

The results showed that **classic bikes** were the most frequently used bike type overall and were preferred by both annual members and casual riders. However, clear differences emerged in the adoption of the remaining bike types. Annual members accounted for a larger proportion of rides across all bike categories, reflecting their higher overall usage of the bike-share system. Casual riders, while fewer in number, demonstrated relatively greater use of electric bikes, suggesting a preference for the additional convenience and reduced physical effort they provide during longer recreational trips.

Ride duration analysis further reinforced these behavioral differences. Regardless of bike type, casual riders consistently recorded longer average trip durations than annual members, indicating that the distinction between the two rider groups is driven more by **trip purpose** than by the bicycle itself. Annual members generally used each bike type for shorter, routine journeys, whereas casual riders tended to undertake longer leisure-oriented rides.

Overall, the bike type analysis indicates that **both rider groups favor classic bikes**, but **their usage patterns differ substantially**. Annual members primarily use bicycles for frequent, practical transportation, while casual riders are more inclined toward longer recreational trips and exhibit a comparatively stronger preference for electric bikes. These findings provide additional context for understanding rider behavior and support future marketing strategies aimed at encouraging casual riders to transition to annual memberships.

- Which stations and routes experience the highest rider activity?

To identify the most frequently used locations within the Cyclistic network, ride activity was analyzed at both the **station** and **route** levels. The analysis examined the most popular start stations, end stations, and origin–destination routes to determine where rider demand was most heavily concentrated and to understand overall travel patterns across the bike-share system.

The results showed that rider activity was concentrated around a relatively small number of high-demand stations. Several locations consistently ranked among the busiest as both trip origins and destinations, indicating that they function as major transportation hubs within the network. The most frequently used routes connected these high-traffic stations, suggesting that riders regularly travel between key commercial, residential, and recreational areas.

Analyzing station usage also revealed that demand was not evenly distributed throughout the network. Instead, a subset of stations generated a disproportionately large share of all trips, while many others experienced comparatively lower activity. This concentration highlights the importance of strategically managing bicycle availability and station capacity at high-demand locations to maintain service reliability.

Overall, the station and route analysis demonstrates that **Cyclistic usage is centered around a core network of highly active stations and travel corridors**. Identifying these demand hotspots provides valuable operational insight for bicycle redistribution, station planning, and infrastructure investment, while also supporting marketing initiatives by revealing where rider engagement is naturally highest.

- How do station preferences differ between annual members and casual riders?

To compare spatial riding behavior, station usage was analyzed separately for annual members and casual riders. The analysis examined the most frequently used start and end stations for each rider type, allowing differences in location preferences to be identified independently of overall ride volume.

The results showed that annual members generated higher ride volumes across most stations, reflecting their greater overall use of the Cyclistic system. Their most frequently used stations were concentrated around major business districts, transit hubs, and densely populated urban areas, indicating that many member trips were associated with routine commuting and everyday transportation.

Casual riders, while using many of the same high-demand stations, displayed stronger preferences for stations located near parks, waterfronts, tourist attractions, and recreational destinations. This suggests that casual riders are more likely to begin and end trips in locations associated with leisure activities rather than daily commuting.

Despite these differences, several stations ranked highly for both rider groups, demonstrating that certain locations serve as central hubs regardless of trip purpose. However, the relative popularity of individual stations varied between annual members and casual riders, reflecting their distinct travel motivations and riding patterns.

Overall, the station preference analysis reinforces the broader findings of the project: **annual members primarily use the bike-share system for practical, transportation-oriented travel, whereas casual riders**

exhibit stronger preferences for recreational destinations and leisure-focused routes. Understanding these spatial differences provides valuable insight for targeted marketing, bicycle redistribution strategies, and future infrastructure planning.

Business Insights

- What key trends and relationships were discovered throughout the analysis?

The analysis revealed several consistent trends that collectively explain how annual members and casual riders use the Cyclistic bike-share system. Across every analytical dimension, including rider composition, ride duration, temporal behavior, bike preferences, and station usage, the two rider groups demonstrated distinctly different usage patterns.

Annual members accounted for the majority of all rides and exhibited consistent riding behavior throughout the year, week, and day. Their trips were characterized by higher ride frequency, shorter average durations, pronounced weekday and commuting-hour activity, and frequent use of stations located near business districts and transit hubs. These patterns indicate that members primarily use Cyclistic as a reliable mode of everyday transportation.

In contrast, casual riders completed fewer trips but consistently recorded longer ride durations. Their activity was more strongly influenced by seasonality, weekends, and leisure hours, with greater engagement during warmer months and at stations near recreational and tourist destinations. Casual riders also demonstrated a comparatively stronger preference for electric bikes, suggesting a greater emphasis on convenience during longer recreational rides.

The ride duration analysis further revealed that trip lengths followed a right-skewed distribution, indicating that a relatively small number of long rides increased the overall average. As a result, distribution-based analyses provided a more representative understanding of rider behavior than average ride duration alone.

Taken together, these findings establish a clear relationship between rider type and trip purpose. Annual members exhibit stable, transportation-oriented behavior driven by routine travel, whereas casual riders display more variable, recreation-oriented behavior influenced by time, season, and location. These consistent behavioral distinctions formed the foundation for the business recommendations developed in the subsequent Share and Act stages.

- How do these analytical findings answer the project's primary business question?

The primary business question sought to understand **how annual members and casual riders differ in their riding behavior** in order to support strategies for converting casual riders into annual members. The analytical findings provide a clear and evidence-based answer by demonstrating that the two rider groups consistently exhibit distinct usage patterns across multiple dimensions of the bike-share system.

The analysis showed that annual members generate the majority of rides and display stable, transportation-oriented behavior characterized by frequent, shorter trips concentrated on weekdays, during commuting hours, and around business districts and transit hubs. In contrast, casual riders take fewer but longer trips, exhibit stronger seasonal and weekend activity, favor recreational locations, and show greater use of electric bikes. These differences remained consistent across temporal, behavioral, and spatial analyses, indicating that the observed patterns reflect fundamentally different travel purposes rather than isolated trends.

By identifying these behavioral distinctions, the analysis provides a data-driven understanding of **who** the two rider groups are, **when** they ride, **where** they ride, and **how** they use the Cyclistic system. This comprehensive understanding directly addresses the project's primary business question and establishes the analytical foundation for the Share stage, where these findings are communicated through visualizations, and the Act stage, where targeted recommendations are developed to encourage casual riders to transition to annual memberships.

Data Project Questions & Considerations

APPASA: Share Stage

Visualization Planning

- What communication objectives guided the dashboard design?

The primary communication objective was to transform the analytical findings into a clear, executive-friendly dashboard that effectively answers the project's business question: **how annual members and casual riders use Cyclistic bikes differently**. The dashboard was designed to present key performance indicators, rider behavior patterns, station usage, and actionable business recommendations in a logical sequence that supports informed decision-making.

Rather than displaying all analyses on a single page, the dashboard was structured to progressively guide stakeholders from a high-level overview to detailed behavioral insights, geographic patterns, and strategic recommendations. This approach enables executives to quickly understand the overall performance while providing sufficient detail to explore the underlying factors driving rider differences.

Interactive slicers for bike type, month, day of week, and rider type were incorporated to allow stakeholders to explore the data dynamically without overwhelming the report. Consistent formatting, colour schemes, and chart selection were also used to improve readability, highlight key comparisons, and ensure that the dashboard communicates insights efficiently to a non-technical business audience.

- How was the dashboard structured to communicate the analysis?

The dashboard was structured as a **four-page analytical narrative** that progressively guides stakeholders from a high-level performance overview to detailed behavioural insights, location-based analysis, and actionable business recommendations. This sequential design ensures that each page builds upon the previous one, allowing decision-makers to understand not only *what* differences exist between annual members and casual riders, but also *why* those differences matter from a business perspective.

The first page, **Executive Overview**, presents key performance indicators and high-level trends to establish the overall ridership landscape. The second page, **Rider Behavior Analysis**, explores behavioural differences through ride patterns, bike preferences, and ride duration distributions. The third page, **Ride Location Analysis**, identifies the most frequently used stations and compares location preferences between rider groups to highlight geographic usage patterns. Finally, the **Strategic Recommendations** page synthesizes the analytical findings into practical, data-driven business recommendations and identifies opportunities to convert casual riders into annual members.

This structured flow enables stakeholders to move naturally from understanding the current state of the business to interpreting rider behaviour and ultimately evaluating evidence-based recommendations that support Cyclistic's membership growth strategy.

- Why was a multi-page dashboard selected instead of a single report page?

A multi-page dashboard was selected to organize the analysis into logical, focused sections while maintaining clarity and readability for stakeholders. Attempting to present all key performance indicators, rider behavior analyses, station insights, and business recommendations on a single page would have resulted in a cluttered report, making it more difficult for users to identify important findings and interpret the analysis effectively.

Each dashboard page was designed to serve a distinct analytical purpose. The **Executive Overview** provides a high-level summary of Cyclistic's overall performance, the **Rider Behavior Analysis** examines behavioural differences between annual members and casual riders, the **Ride Location Analysis** explores station-level usage patterns, and the **Strategic Recommendations** page translates the analytical findings into actionable business initiatives. This separation allows each topic to be explored in sufficient detail without overwhelming the audience.

Organizing the dashboard into multiple pages also creates a natural storytelling flow, enabling stakeholders to progress from summary metrics to detailed analysis and ultimately to evidence-based recommendations. This structure improves the overall user experience, supports more effective decision-making, and aligns with dashboard design best practices by presenting information in manageable, logically connected sections.

- How were visualizations mapped to the project's business question?

The visualizations were intentionally selected to address different aspects of the project's primary business question: **how annual members and casual riders use Cyclistic bikes differently**. Rather than presenting unrelated charts, each visualization was designed to examine a specific dimension of rider behavior and collectively provide a comprehensive understanding of the differences between the two rider groups.

The **Executive Overview** page uses KPI cards and trend charts to establish high-level differences in ride volume, average ride length, monthly ridership trends, hourly activity, and weekly usage patterns. These visuals provide an overall comparison of rider behaviour before exploring more detailed analyses.

The **Rider Behavior Analysis** page focuses on behavioural characteristics by examining ride activity throughout the day, bike type preferences, and ride duration distributions. Where the objective was to compare behavioural composition independent of rider volume, percentage-based visualizations were used. Conversely, when analysing actual demand patterns, raw ride counts were presented to preserve the magnitude of rider activity.

The **Ride Location Analysis** page investigates spatial usage patterns by identifying the busiest start and end stations, comparing station usage between rider groups, and highlighting stations associated with the longest average ride durations. These visualizations reveal where rider behaviour differs geographically and support operational and marketing insights.

Finally, the **Strategic Recommendations** page synthesizes the findings from the previous analyses into evidence-based business recommendations. Rather than introducing additional charts, it communicates how the analytical insights can be applied to develop targeted strategies for converting casual riders into annual members.

By assigning each visualization to a specific analytical objective, the dashboard maintains a clear narrative flow and ensures that every visual directly contributes to answering the project's business question.

- Why was Power BI selected as the visualization platform?

Power BI was selected as the visualization platform because it provides a comprehensive business intelligence environment for transforming processed data into interactive, professional dashboards that support data-driven decision-making. Its robust data modeling capabilities, interactive visualizations, and user-friendly interface make it well suited for communicating analytical findings to both technical and non-technical stakeholders.

The processed dataset generated during the Process stage was imported into Power BI, where relationships, calculated measures, and a dedicated date table were created to support dynamic analysis. Interactive features such as slicers and cross-filtering enable users to explore rider behaviour by rider type, bike type, month, and day of the week without modifying the underlying dataset, making the dashboard flexible and easy to navigate.

Power BI also offers a wide range of customizable visualizations that effectively communicate trends, comparisons, distributions, and geographic usage patterns. These capabilities allowed the dashboard to present complex analytical findings in a clear and accessible manner while maintaining a consistent design across all report pages.

Overall, Power BI was chosen because it combines data modeling, interactive exploration, and high-quality visualization within a single platform, enabling the analytical results to be translated into meaningful business insights and actionable recommendations for Cyclistic's stakeholders.

Visualization Design & Dashboard Architecture

- Why were these visualization types selected?

The visualization types were selected based on the nature of the data and the analytical objective of each section, ensuring that every visual communicated insights clearly and supported meaningful comparisons between annual members and casual riders. Rather than choosing charts for visual appeal, each visualization was matched to the type of information being presented and the business question it was intended to answer.

KPI cards were used to highlight key metrics such as total rides, average ride duration, and rider counts, allowing stakeholders to quickly assess overall performance. **Line charts** were selected to display monthly, weekly, and hourly trends because they effectively illustrate changes in rider activity over time. **Column and bar charts** were used to compare categorical variables, including bike type preferences and station usage, making it easy to identify differences in rider behaviour across categories. **Percentage-based stacked bar charts** were chosen where the objective was to compare the relative distribution of rider preferences rather than absolute ride volumes. **Matrix heatmaps** were incorporated to visualize rider activity across combinations of days and hours, enabling patterns of peak usage to be identified efficiently.

Each visualization was designed to answer a specific analytical question while complementing the others within the dashboard. Collectively, the selected visuals simplify complex datasets, emphasize the differences between rider groups, and communicate findings in a format that supports informed business decision-making.

- How do the chosen visualizations support comparisons between annual members and casual riders?

The dashboard visualizations were specifically designed to facilitate direct comparisons between annual members and casual riders across multiple dimensions of rider behaviour. Consistent colour coding, standardized chart formats, and side-by-side comparisons were used throughout the dashboard to help stakeholders quickly identify similarities, differences, and trends between the two rider groups.

Charts comparing ride volume, average ride duration, hourly activity, weekly riding patterns, bike type preferences, and station usage display both rider groups within the same visual, enabling meaningful comparisons without requiring users to switch between separate charts. This design makes differences in riding behaviour immediately apparent while maintaining consistency across the dashboard.

Where comparisons focused on rider composition or preference, **percentage-based visualizations** were used to normalize the data and eliminate the influence of differing ride volumes between the two groups. Conversely, when the objective was to evaluate operational demand or overall usage, **absolute ride counts** were presented to accurately reflect the scale of rider activity.

Interactive slicers further enhance comparisons by allowing stakeholders to filter both rider groups simultaneously across dimensions such as month, bike type, and day of the week. This enables users to investigate how behavioural differences change under different conditions while preserving a consistent comparison framework.

By combining consistent visual design, appropriate chart selection, normalized metrics where necessary, and interactive filtering, the dashboard provides a comprehensive and intuitive comparison of annual member and casual rider behaviour, directly supporting Cyclistic's objective of identifying opportunities to increase annual memberships.

- Which visualizations communicate proportions rather than raw counts, and why?

Percentage-based visualizations were used whenever the objective was to compare the relative behaviour and preferences of annual members and casual riders rather than the total number of rides. Since annual members generated substantially more trips than casual riders, relying solely on raw ride counts could have obscured meaningful behavioural differences by emphasizing differences in rider volume instead of rider characteristics.

In the dashboard, 100% stacked bar charts were used to compare variables such as bike type preferences and ride duration distributions. These visualizations normalize the data by expressing each category as a percentage of the total rides within each rider group, allowing stakeholders to compare behavioural patterns independently of group size.

Using proportions makes it easier to identify differences in preferences. For example, it highlights whether casual riders have a greater tendency toward longer ride durations or different bike type choices relative to annual members, regardless of the fact that members complete more rides overall. This provides a fair and meaningful comparison of rider behaviour.

In contrast, visualizations intended to represent operational demand, such as monthly ridership trends, station usage, and total ride volume, were displayed using raw counts because the actual magnitude of rider activity is essential for understanding usage patterns and supporting business decisions. By combining proportional and count-based visualizations where appropriate, the dashboard communicates both relative behavioural differences and overall rider demand in a balanced and informative manner.

- How were KPI metrics selected?

The KPI metrics were selected to provide an immediate summary of the performance indicators most relevant to the project's business question: **how annual members and casual riders differ in their usage of Cyclistic bikes**. Each KPI was chosen because it represents a fundamental measure of rider behaviour and serves as a starting point for the more detailed analyses presented throughout the dashboard.

The selected KPIs include **Total Rides**, **Average Ride Duration**, **Annual Member Rides**, **Casual Rider Rides**, **Average Ride Duration for Annual Members**, and **Average Ride Duration for Casual Riders**. Together, these metrics provide a concise overview of overall ridership, the distribution of riders between membership types, and differences in ride duration, enabling stakeholders to quickly understand the scale and nature of rider behaviour.

The KPIs were also selected to complement the supporting visualizations rather than replace them. While the KPI cards present high-level summary metrics, the accompanying charts explain the underlying trends, behavioural patterns, and geographic differences that contribute to those values. This combination allows users to move naturally from summary statistics to detailed analysis without losing context.

By focusing on metrics that directly align with the project's objectives, the KPI section provides an effective executive summary and establishes a clear foundation for interpreting the analytical findings presented in the rest of the dashboard.

- How were filters and slicers incorporated into the dashboard?

Interactive filters and slicers were incorporated throughout the dashboard to enable stakeholders to explore the data dynamically while maintaining a consistent analytical framework. Rather than creating multiple static reports, slicers allow users to customize the displayed information based on specific categories and immediately observe how rider behaviour changes across different conditions.

The dashboard includes slicers for **rider type**, **bike type**, **month**, and **day of the week**, allowing users to filter visualizations according to their analytical needs. These slicers are synchronized across the relevant report pages so that selected filters are applied consistently to all associated visualizations, ensuring meaningful comparisons and reducing the need for repetitive user interactions.

Power BI's cross-filtering functionality further enhances the dashboard by allowing interactions between visualizations. Selecting a category or data point within one chart automatically updates the related charts on the page, enabling users to investigate relationships between different aspects of rider behaviour without modifying the underlying dataset.

By incorporating interactive filters and slicers, the dashboard becomes more flexible and user-centric, allowing stakeholders to explore seasonal patterns, bike preferences, and rider characteristics from multiple perspectives while preserving a clear and consistent presentation of the analytical findings.

- Why were certain insights displayed as KPI cards instead of charts?

KPI cards were used for insights that represent **high-level summary metrics** rather than trends, distributions, or comparisons. Their purpose is to provide stakeholders with an immediate overview of the dashboard's most important performance indicators before they examine the detailed visualizations.

Metrics such as **Total Rides**, **Average Ride Duration**, **Annual Member Rides**, **Casual Rider Rides**, **Average Ride Duration for Annual Members**, and **Average Ride Duration for Casual Riders** are single aggregated values that can be understood quickly without graphical representation. Displaying these metrics as KPI cards reduces visual complexity and enables users to identify the overall scale and composition of rider activity at a glance.

Charts were reserved for analyses involving relationships, comparisons, and trends across multiple categories or time periods, where graphical representation provides greater analytical value. For example, line charts illustrate changes in ridership over time, bar charts compare bike type preferences and station usage, and heatmaps reveal patterns of rider activity across days and hours. These visualizations communicate information that cannot be effectively represented by a single numeric value.

Using KPI cards alongside charts creates a clear information hierarchy within the dashboard. The KPI cards provide a concise executive summary, while the accompanying visualizations explain the underlying patterns and factors that contribute to those metrics. This combination improves readability, supports faster decision-making, and allows stakeholders to transition naturally from high-level performance indicators to detailed behavioural analysis.

- Why was the dashboard organized into four sequential pages, and how does this structure guide users through the analysis?

The dashboard was organized into four sequential pages to present the analysis as a logical business narrative rather than a collection of independent visualizations. Each page has a distinct analytical purpose and builds upon the insights from the previous page, enabling stakeholders to progressively understand the data, identify behavioural patterns, and arrive at evidence-based business recommendations. This structured approach improves readability, reduces information overload, and supports informed decision-making.

The dashboard begins with the **Executive Overview**, which provides a concise summary of key performance indicators, overall ridership trends, and high-level comparisons between annual members and casual riders. This page establishes the analytical context and allows stakeholders to quickly understand the scale and composition of the dataset before exploring more detailed findings.

The second page, **Rider Behavior Analysis**, investigates how the two rider groups differ in their riding habits. Visualizations examine ride duration, hourly and weekly usage patterns, and bike type preferences, revealing behavioural differences that directly address the project's business question.

The third page, **Ride Location Analysis**, extends the analysis by examining spatial usage patterns, including the most frequently used stations and location-based rider behaviour. This page provides geographic context that helps explain where demand is concentrated and how rider groups utilize the bike-share network differently.

The final page, **Strategic Recommendations**, synthesizes the analytical findings into actionable business recommendations. Rather than introducing new analyses, this page connects the evidence presented throughout the dashboard to practical strategies aimed at increasing annual memberships, ensuring that stakeholders can clearly understand how the analysis supports the proposed recommendations.

By following this sequential structure, moving from the executive summary to behavioural analysis, then to geographic insights, and finally to strategic recommendations, the dashboard guides users through a coherent analytical journey. Each page answers a specific set of questions while naturally leading to the next stage of analysis, resulting in a clear, evidence-based narrative that effectively supports business decision-making.

- How do the dashboard pages work together to communicate the complete analytical story?

The four dashboard pages were designed to function as a connected analytical narrative, with each page contributing a distinct layer of insight while building upon the findings of the previous one. Rather than presenting isolated visualizations, the dashboard guides stakeholders through a structured journey from understanding the overall performance of the bike-share system to identifying behavioural patterns and ultimately arriving at data-driven business recommendations.

The **Executive Overview** establishes the foundation by summarizing the key performance indicators and presenting an overview of ridership trends. This page answers the question, *"What does the overall data look like?"* and provides the context needed for the subsequent analyses.

Building on this foundation, the **Rider Behavior Analysis** page examines *how* annual members and casual riders differ in their riding habits. By exploring ride duration, riding frequency, bike type preferences, and temporal usage patterns, this page identifies the behavioural differences that explain the high-level metrics presented in the Executive Overview.

The **Ride Location Analysis** page then adds a geographic perspective by showing *where* these behaviours occur. Analysing station usage and location-based ride patterns helps stakeholders understand how spatial demand varies between rider groups and reveals opportunities for targeted operational or marketing strategies.

Finally, the **Strategic Recommendations** page synthesizes the evidence gathered throughout the previous pages into actionable business recommendations. Instead of introducing new analyses, it links the behavioural and geographic insights directly to strategies for converting casual riders into annual members, ensuring that every recommendation is supported by the findings presented earlier in the dashboard.

Together, these four pages create a cohesive analytical story that progresses from **what happened**, to **how riders behave**, to **where those behaviours occur**, and ultimately to **what actions Cyclistic should take**. This logical sequence enables stakeholders to understand not only the analytical findings but also how those findings support informed, evidence-based business decisions.

Business Storytelling & Communication

- Who is the intended audience for the dashboard?

The dashboard was designed primarily for **Cyclistic's executive leadership, marketing team, and business decision-makers** who are responsible for developing strategies to increase annual memberships. As these stakeholders may have varying levels of technical expertise, the dashboard emphasizes clear visual communication, intuitive navigation, and concise presentation of key findings rather than complex statistical analysis.

For executives, the dashboard provides high-level KPI metrics and summary visualizations that support rapid assessment of overall rider performance and membership trends. For marketing and business teams, the detailed behavioural and geographic analyses offer actionable insights into how casual riders differ from annual members, helping identify opportunities for targeted campaigns, promotional offers, and customer engagement strategies.

The dashboard is also suitable for analysts and project stakeholders who require a deeper understanding of the underlying data. Interactive filters, detailed visualizations, and supporting comparisons allow users to explore rider behaviour across different dimensions while maintaining a consistent analytical framework.

By balancing executive-level summaries with detailed analytical insights, the dashboard effectively serves both strategic decision-makers and analytical users, ensuring that the findings are accessible, actionable, and aligned with Cyclistic's objective of converting more casual riders into annual members.

- How does the dashboard answer the original business question?

The dashboard was designed specifically to answer the project's central business question: **How do annual members and casual riders use Cyclistic bikes differently?** It accomplishes this by presenting a comprehensive comparison of the two rider groups across multiple behavioural, temporal, and geographic dimensions, allowing stakeholders to identify the patterns that distinguish casual riders from annual members.

The **Executive Overview** establishes the overall context by summarizing key performance indicators and ridership trends, highlighting differences in rider composition and average ride duration. The **Rider Behavior Analysis** then explores how usage varies between the two groups by examining ride duration, hourly activity, weekday usage, and bike type preferences. These visualizations reveal that annual members primarily use the service for regular weekday commuting with shorter, more consistent trips, while casual riders tend to take longer recreational rides, particularly during weekends and leisure periods.

The **Ride Location Analysis** complements these findings by identifying where riders begin and end their journeys, revealing location-based differences in travel behaviour and station usage. These geographic insights provide additional context for understanding how and where each rider group interacts with the bike-share system.

Finally, the **Strategic Recommendations** page connects the analytical findings to Cyclistic's business objective by translating the observed behavioural differences into practical marketing and operational recommendations aimed at encouraging casual riders to become annual members.

By integrating executive summaries, behavioural comparisons, geographic analyses, and evidence-based recommendations into a single analytical workflow, the dashboard provides a complete and data-driven answer to the original business question while directly supporting Cyclistic's goal of increasing annual memberships.

- What story do the visualizations communicate?

The visualizations collectively communicate the story of **how annual members and casual riders differ in their usage of Cyclistic's bike-share service and how these behavioural differences can inform strategies to increase annual memberships**. Rather than presenting isolated charts, the dashboard arranges the visualizations to progressively build an evidence-based narrative that answers the project's business question and supports strategic decision-making.

The story begins by establishing the overall performance of the bike-share system through KPI cards and summary visualizations, providing stakeholders with a clear understanding of rider composition, ride volume, and average ride duration. It then transitions into a detailed comparison of rider behaviour, illustrating that annual members primarily use the service for consistent weekday commuting with shorter rides, while casual riders tend to take longer, recreational trips that are concentrated during weekends and leisure hours.

The narrative is further developed through geographic visualizations, which demonstrate where rider activity is concentrated and highlight differences in station usage between the two rider groups. These spatial insights complement the behavioural analysis by revealing how location influences rider patterns and identifying areas where marketing or operational initiatives could be most effective.

Finally, the dashboard concludes by connecting these findings to practical business recommendations. The visualizations collectively demonstrate that the behavioural differences between casual riders and annual members present clear opportunities for targeted membership campaigns, seasonal promotions, and location-specific marketing strategies aimed at encouraging casual riders to transition to annual memberships.

By progressing from **overall performance**, to **behavioural differences**, to **geographic insights**, and ultimately to **actionable recommendations**, the dashboard communicates a complete analytical story that transforms raw data into meaningful business insights and directly supports Cyclistic's strategic objective of increasing annual memberships.

- How were analytical findings translated into business recommendations?

The analytical findings were translated into business recommendations by identifying consistent behavioural patterns within the data and linking those patterns to Cyclistic's objective of increasing annual memberships. Rather than presenting observations in isolation, the analysis focused on understanding how differences in rider behaviour could be leveraged to develop targeted marketing and operational strategies.

The dashboard revealed that annual members primarily use the bike-share service for regular weekday commuting, while casual riders are more likely to take longer recreational trips during weekends and leisure hours. These behavioural differences indicated that casual riders value flexibility and leisure-oriented experiences, whereas annual members benefit from frequent and routine usage. Based on these insights, recommendations were developed to encourage casual riders to recognize the long-term value of an annual membership.

Geographic analysis further supported the recommendations by identifying stations and locations with high concentrations of casual rider activity. These findings suggest opportunities to implement location-specific promotional campaigns, membership advertisements, and targeted outreach in areas where conversion potential is highest. Seasonal ridership trends and usage patterns also informed recommendations for offering promotional discounts and marketing campaigns during periods of increased casual rider activity.

Each recommendation presented in the dashboard is directly supported by evidence from the behavioural, temporal, and geographic analyses, ensuring that proposed actions are grounded in data rather than assumptions. By translating analytical insights into practical strategies, the dashboard bridges the gap between data analysis and business decision-making, providing Cyclistic with actionable recommendations to support its goal of converting more casual riders into annual members.

- How do the recommendations relate to the analysis?

The recommendations are directly derived from the analytical findings and represent the practical application of the patterns identified throughout the dashboard. Rather than being based on assumptions or general best practices, each recommendation is supported by evidence obtained from the behavioural, temporal, and geographic analyses conducted during the project.

The analysis showed that annual members primarily use Cyclistic bikes for consistent weekday commuting, while casual riders exhibit longer ride durations and higher usage during weekends and leisure periods. These behavioural differences indicate that casual riders interact with the service differently and therefore require marketing strategies tailored to their specific usage patterns. Consequently, the recommendations focus on promoting the long-term value of annual memberships to riders who frequently use the service for recreational purposes.

Similarly, the geographic analysis identified stations and areas with high concentrations of casual rider activity, providing evidence for location-specific marketing initiatives and membership promotions. Seasonal and temporal usage trends also informed recommendations for launching targeted campaigns during periods when casual rider demand is highest, increasing the likelihood of successful membership conversions.

By ensuring that every recommendation can be traced back to one or more analytical findings, the dashboard maintains a clear connection between **data, insight, and action**. This evidence-based approach strengthens the credibility of the proposed strategies and demonstrates how the analysis supports Cyclistic's business objective of converting more casual riders into annual members.

- How does the dashboard support executive decision-making?

The dashboard supports executive decision-making by transforming complex analytical results into a clear, concise, and actionable business report. Rather than requiring stakeholders to interpret raw data or detailed statistical outputs, the dashboard presents the most important findings through intuitive visualizations, executive-level KPI summaries, and evidence-based recommendations that directly address Cyclistic's business objective of increasing annual memberships.

The **Executive Overview** provides an immediate snapshot of key performance indicators, enabling decision-makers to quickly assess overall rider activity, membership composition, and ride duration without reviewing the underlying dataset. Subsequent pages progressively explain the behavioural and geographic factors that contribute to these high-level metrics, allowing executives to understand not only *what* is happening but also *why* these patterns occur.

Interactive filters and synchronized visualizations further support decision-making by allowing stakeholders to explore specific rider segments, time periods, and bike types without compromising the overall analytical narrative. This flexibility enables executives to investigate areas of interest while maintaining consistent comparisons across the dashboard.

Most importantly, the dashboard concludes with a dedicated **Strategic Recommendations** page that translates analytical findings into practical business actions. By linking every recommendation directly to evidence presented in the preceding analyses, the dashboard ensures that strategic decisions are supported by data rather than assumptions. This evidence-based approach helps executives confidently identify opportunities for targeted marketing campaigns, membership promotions, and resource allocation, ultimately supporting Cyclistic's objective of converting more casual riders into annual members.

Accessibility & Presentation

- How was accessibility considered during dashboard design?

Accessibility was an important consideration throughout the dashboard design to ensure that the analytical findings could be easily understood by users with varying levels of technical expertise. The dashboard emphasizes clarity, consistency, and intuitive navigation so that executives, marketing teams, and analysts can quickly interpret the information and make informed decisions.

A consistent colour palette, typography, and formatting were applied across all four dashboard pages to create a cohesive user experience and reduce visual distractions. Charts were selected based on their ability to communicate information clearly, while descriptive titles, axis labels, legends, and data labels were included where appropriate to improve interpretation. Consistent colour coding was also used to represent annual members and casual riders throughout the dashboard, allowing users to identify rider groups immediately without repeatedly consulting legends.

The dashboard follows a logical reading order, progressing from executive-level summaries to detailed behavioural analysis, geographic insights, and strategic recommendations. This structured layout reduces cognitive load and enables stakeholders to understand the analytical story step by step. Interactive slicers were placed in consistent locations and synchronized across pages, ensuring that users could filter the data without disrupting the overall analytical workflow.

Visual clutter was minimized by avoiding unnecessary decorative elements and limiting each page to a focused set of related visualizations. High-contrast text, adequate spacing, and appropriately sized fonts further enhance readability across different screen sizes and presentation settings.

By prioritizing simplicity, consistency, and intuitive navigation, the dashboard ensures that its insights are accessible to both technical and non-technical stakeholders, enabling a wider audience to confidently interpret the findings and support data-driven business decisions.

- How were colours used to improve readability?

Colours were used strategically throughout the dashboard to improve readability, reinforce visual consistency, and help users quickly distinguish between key categories without overwhelming the interface. Rather than using numerous decorative colours, the dashboard follows a restrained and consistent colour scheme that supports effective data interpretation.

A consistent colour palette was applied across all four dashboard pages, with the same colours representing **annual members** and **casual riders** in every visualization. This consistency allows stakeholders to recognize rider groups immediately, reducing the need to repeatedly reference chart legends and making comparisons more intuitive. Neutral background colours and subtle accent colours were used to maintain visual balance while ensuring that the data remained the primary focus.

Colour was also used to establish visual hierarchy. KPI cards, chart titles, and important metrics were highlighted using accent colours to draw attention to key insights, while supporting information was displayed using more neutral tones. This helps users quickly identify the most important information without becoming distracted by less critical details.

High colour contrast between text, backgrounds, and chart elements was maintained to enhance readability and improve accessibility for a wide range of users. The limited use of colours also reduces visual clutter, allowing stakeholders to focus on patterns, trends, and comparisons rather than decorative design elements.

By applying colours consistently and purposefully, the dashboard improves readability, strengthens comparisons between rider groups, and creates a clear, professional interface that effectively communicates analytical insights to both technical and non-technical audiences.

- How were titles, subtitles, and labels designed?

Titles, subtitles, and labels were designed to enhance clarity, provide context, and guide users through the analytical narrative without requiring additional explanation. Each dashboard page includes a descriptive title that clearly communicates its purpose, while concise subtitles provide context about the analyses being presented and their relevance to Cyclistic's business objective.

Chart titles were written using clear, business-oriented language that describes the insight being presented rather than simply naming the visualization. This allows stakeholders to immediately understand the focus of each chart without needing to interpret technical terminology. Where appropriate, subtitles were used to provide additional context, explain the scope of the analysis, or indicate the dimensions being compared.

Axis labels, legends, and data labels were kept concise and consistent throughout the dashboard to improve readability. Standardized terminology was used across all visualizations to ensure that the same metrics, rider categories, and time periods were described consistently, reducing potential confusion. Labels were formatted using readable font sizes and positioned to avoid overlapping with chart elements, ensuring that important information remained visible and easy to interpret.

The naming convention for dashboard pages also follows the logical progression of the analytical story, moving from **Executive Overview** to **Rider Behavior Analysis**, **Ride Location Analysis**, and finally **Strategic Recommendations**. This consistent structure helps users navigate the dashboard intuitively while reinforcing the flow from summary metrics to business recommendations.

By using clear, descriptive, and consistently formatted titles, subtitles, and labels, the dashboard improves usability, minimizes ambiguity, and enables both technical and non-technical stakeholders to quickly understand and interpret the analytical findings.

- What design choices reduce cognitive load?

Several design choices were incorporated to reduce cognitive load and help users interpret the dashboard efficiently. The dashboard follows a clear and consistent layout that presents information in a logical sequence, allowing stakeholders to focus on understanding the insights rather than navigating the interface. By dividing the analysis into four dedicated pages, **Executive Overview**, **Rider Behavior Analysis**, **Ride Location Analysis**, and **Strategic Recommendations**, the dashboard avoids overwhelming users with excessive information on a single screen.

Each page contains only a focused set of related visualizations, ensuring that users can concentrate on one aspect of the analysis at a time. Consistent placement of KPI cards, charts, slicers, and navigation elements further reduces the effort required to locate information, while synchronized filters allow users to explore different perspectives without learning new interactions on each page.

Visual consistency also contributes to reducing cognitive load. The dashboard uses standardized colours, typography, chart styles, and terminology across all visualizations, enabling users to recognize recurring patterns and comparisons quickly. Clear titles, concise labels, and descriptive legends provide sufficient context without adding unnecessary detail, while unnecessary decorative elements were intentionally avoided to keep attention on the data.

An intuitive visual hierarchy ensures that the most important information is seen first. Executive-level KPI cards are positioned prominently, followed by supporting charts that explain the underlying trends and relationships. This progression allows stakeholders to move naturally from summary metrics to detailed analysis and finally to strategic recommendations.

By emphasizing simplicity, consistency, and a structured analytical flow, the dashboard minimizes cognitive effort and enables both technical and non-technical stakeholders to interpret the findings quickly and make informed, data-driven decisions.

- Why was consistent formatting important?

Consistent formatting was essential to create a professional, easy-to-navigate dashboard that enables stakeholders to interpret information quickly and accurately. By applying the same design standards across all four dashboard pages, users can focus on understanding the analytical findings rather than adapting to different layouts or visual styles.

Uniform formatting was maintained through the consistent use of colours, typography, chart styles, KPI card designs, spacing, and page layouts. The same colour scheme was used to represent annual members and casual riders throughout the dashboard, while standardized fonts, titles, and labels ensured that similar information was presented in a familiar way across every visualization. This consistency reduces confusion and makes comparisons between charts more intuitive.

Consistent formatting also reinforces the dashboard's analytical narrative. As users progress from the **Executive Overview** to **Rider Behavior Analysis**, **Ride Location Analysis**, and **Strategic Recommendations**, they encounter a familiar visual structure that supports a smooth and logical flow of information. This predictable layout minimizes cognitive effort and allows stakeholders to concentrate on the insights being communicated.

From a business perspective, consistent formatting enhances the dashboard's credibility and professionalism. A well-organized and visually cohesive report improves stakeholder confidence in the analysis and facilitates effective communication during presentations and decision-making discussions.

By maintaining a consistent visual design throughout the dashboard, the analysis becomes easier to navigate, comparisons become more meaningful, and the overall user experience is significantly improved for both technical and non-technical audiences.

- Why was a dedicated Strategic Recommendations page included?

A dedicated **Strategic Recommendations** page was included to bridge the gap between data analysis and business decision-making. While the preceding dashboard pages focus on presenting and explaining analytical findings, the final page translates those insights into clear, actionable recommendations that directly support Cyclistic's objective of increasing annual memberships.

The dashboard progressively builds an analytical narrative by first summarizing overall performance, then examining rider behaviour and geographic usage patterns. The Strategic Recommendations page serves as the conclusion to this narrative by synthesizing the evidence from the previous analyses and demonstrating how the identified patterns can be used to inform marketing and operational strategies. This ensures that stakeholders are not only presented with data but are also provided with practical guidance on how to act upon the findings.

Separating the recommendations from the analytical visualizations also improves readability and prevents the dashboard from becoming cluttered. Decision-makers can review the supporting evidence throughout the earlier pages before referring to a concise summary of recommended actions, creating a logical progression from **data**, to **insight**, and finally to **business action**.

Each recommendation on this page is explicitly linked to findings from the behavioural, temporal, and geographic analyses, reinforcing that the proposed strategies are evidence-based rather than speculative. By presenting actionable insights in a dedicated section, the dashboard enables executives and marketing teams to quickly identify opportunities for targeted campaigns, membership promotions, and resource allocation.

Including a separate Strategic Recommendations page therefore enhances the dashboard's value as a business decision-support tool, ensuring that the analysis not only answers the original business question but also provides a clear path for achieving Cyclistic's strategic goal of converting more casual riders into annual members.

- How does the dashboard guide the audience from findings to recommendations?

The dashboard guides the audience from findings to recommendations by following a structured analytical narrative that progressively transforms raw data into actionable business insights. Rather than presenting visualizations as isolated analyses, each dashboard page builds upon the previous one, enabling stakeholders to understand not only the results but also their business implications.

The journey begins with the **Executive Overview**, which provides a concise summary of key performance indicators and overall ridership trends, establishing the context for the analysis. The **Rider Behavior Analysis** then explores the differences between annual members and casual riders by examining ride duration, hourly activity, weekday usage, and bike type preferences. These behavioural insights explain how the two rider groups interact with the bike-share service and identify the patterns that distinguish them.

The **Ride Location Analysis** extends the narrative by identifying where rider activity is concentrated and highlighting location-based differences in station usage. This additional layer of analysis provides geographic context, helping stakeholders understand where marketing efforts and operational improvements could have the greatest impact.

Finally, the **Strategic Recommendations** page synthesizes the evidence presented throughout the dashboard into clear, practical actions. Each recommendation is explicitly supported by findings from the behavioural, temporal, and geographic analyses, demonstrating how the observed patterns can be translated into targeted marketing campaigns, membership promotions, and operational strategies.

By progressing logically from **executive summary**, to **behavioural insights**, to **geographic analysis**, and ultimately to **business recommendations**, the dashboard creates a coherent flow from **data**, to **evidence**, to **decision-making**. This structured approach ensures that stakeholders can clearly understand how each recommendation is supported by the analysis, making the proposed actions both credible and actionable.

Data Project Questions & Considerations

APPASA: Act Stage

Business Conclusions

- What is the overall conclusion of this analysis?

The analysis concludes that annual members and casual riders exhibit fundamentally different usage patterns, reflecting distinct transportation needs and motivations. Annual members primarily use Cyclistic bicycles for routine commuting, characterized by shorter ride durations, consistent weekday usage, and pronounced peak-hour travel. In contrast, casual riders demonstrate longer, leisure-oriented trips, stronger weekend activity, and increased ridership during the warmer months.

These behavioural differences indicate that casual riders represent the greatest opportunity for membership growth. Rather than attempting to alter rider behaviour, Cyclistic should align its marketing strategy with existing recreational usage patterns by promoting membership options and incentives that appeal to high-engagement casual riders. By targeting this segment with tailored value propositions and flexible membership offerings, Cyclistic can increase annual memberships while leveraging established rider behaviour patterns.

- Which findings have the greatest business impact?

Three findings have the greatest business impact because they directly inform strategies for increasing annual memberships. First, casual riders consistently take longer, leisure-oriented trips and are disproportionately represented in extended ride duration categories, indicating that they primarily use Cyclistic for recreation rather than routine transportation. Second, annual members exhibit predictable weekday commuting patterns with shorter ride durations and pronounced peak-hour usage, demonstrating that memberships currently deliver the greatest value to regular commuters. Third, casual rider activity increases substantially during weekends and the warmer months, creating clear seasonal opportunities to engage highly active riders when they are most likely to consider membership.

Together, these findings suggest that the most effective growth strategy is not to change how casual riders use the service, but to design membership offerings and marketing campaigns that align with their existing recreational riding behaviour. This insight provides a clear, evidence-based direction for Cyclistic to increase annual memberships by targeting high-engagement casual riders with tailored promotional strategies and flexible membership options.

Business Recommendations

- Why was promoting leisure-oriented membership value recommended?

Promoting leisure-oriented membership value was recommended because the analysis demonstrated that casual riders primarily use Cyclistic bicycles for recreational purposes rather than routine commuting. Casual riders consistently recorded longer average ride durations, were more heavily represented in extended ride duration categories, and exhibited stronger weekend and seasonal riding patterns. These behavioural characteristics indicate that their motivations differ substantially from those of annual members.

As a result, marketing memberships primarily as a commuting solution is unlikely to resonate with this audience. Instead, emphasizing benefits that support recreational riding, such as extended ride value, weekend-focused rewards, scenic route campaigns, and leisure-oriented promotions, better aligns the membership offering with existing rider behaviour. By positioning annual memberships as a valuable option for frequent recreational riders, Cyclistic can increase the attractiveness of membership without requiring casual riders to change how they use the service.

- Why was reducing commitment barriers recommended?

Reducing commitment barriers was recommended because the analysis indicates that many casual riders use Cyclistic frequently enough to represent strong membership prospects but may be discouraged by the perceived commitment of purchasing an annual membership. Their recurring recreational usage suggests continued engagement with the service, yet the preference for single-ride or day-pass options implies that the existing membership structure may not adequately address their needs or purchasing preferences.

Introducing lower-commitment membership options, trial memberships, limited-time introductory offers, or promotional incentives could encourage casual riders to experience the benefits of membership with reduced perceived risk. By lowering the barriers to entry, Cyclistic can increase the likelihood that engaged casual riders transition to annual memberships while supporting the company's objective of expanding its long-term member base.

- Why were seasonal and peak conversion campaigns recommended?

Seasonal and peak conversion campaigns were recommended because the analysis revealed that casual rider activity increases significantly during weekends and the warmer months, when recreational cycling demand is highest. These periods represent times of increased engagement with the Cyclistic service, making casual riders more receptive to membership-related marketing and promotional offers.

By concentrating marketing efforts during these high-activity periods, Cyclistic can reach riders when they are experiencing the greatest value from the service and are more likely to recognize the benefits of an annual membership. Aligning campaigns with periods of peak usage enables the company to maximize marketing effectiveness, improve conversion opportunities, and support its objective of increasing annual memberships through timely, data-driven customer engagement.

- How do the recommendations support Cyclistic's objective of increasing annual memberships?

The recommendations directly support Cyclistic's objective of increasing annual memberships by aligning marketing strategies with the observed behaviour and preferences of casual riders. Rather than applying broad, generic promotions, each recommendation addresses a specific behavioural pattern identified during the analysis. Promoting leisure-oriented membership value aligns with the recreational usage patterns of casual riders, reducing commitment barriers makes membership more accessible to engaged riders who may be hesitant to commit to an annual plan, and seasonal conversion campaigns target periods when casual rider activity and engagement are at their highest.

Together, these recommendations provide a targeted, data-driven approach to converting existing casual riders into annual members. By focusing on the rider segment with the greatest conversion potential and tailoring marketing initiatives to their demonstrated behaviour, Cyclistic can improve the effectiveness of its membership campaigns and support sustainable long-term growth.

- How are the recommendations supported by the analytical findings?

The recommendations are directly supported by the analytical findings identified throughout the analysis. The recommendation to promote leisure-oriented membership value is supported by evidence that casual riders primarily engage in longer, recreational trips and demonstrate stronger weekend usage patterns. Reducing commitment barriers is supported by the finding that casual riders use the service regularly but continue to rely on single-ride or short-term purchase options, suggesting an opportunity to encourage membership through more accessible entry points. Seasonal and peak conversion campaigns are supported by the observed increase in casual rider activity during the warmer months and periods of higher recreational demand.

Because each recommendation is derived from measurable behavioural trends rather than assumptions, they provide an evidence-based strategy for increasing annual memberships. This alignment between analytical findings and business recommendations ensures that Cyclistic's marketing initiatives address the characteristics and motivations of the rider segment with the greatest conversion potential.

Business Impact & Next Steps

- How can Cyclistic apply these insights?

Cyclistic can apply these insights by developing marketing strategies and membership offerings that reflect the behavioural patterns identified throughout the analysis. Rather than using a one-size-fits-all approach, the company should focus its efforts on high-engagement casual riders by promoting membership benefits that align with recreational cycling, introducing lower-commitment membership options, and launching targeted campaigns during weekends and peak riding seasons when casual rider activity is highest.

In addition, Cyclistic can use these insights to guide future marketing decisions, allocate promotional resources more effectively, and continuously evaluate campaign performance using key metrics such as membership conversion rates, customer acquisition costs, and member retention. Applying data-driven strategies based on observed rider behaviour enables the company to improve the effectiveness of its marketing initiatives while supporting the long-term objective of increasing annual memberships.

- What actions should the marketing team take next?

The marketing team should develop and implement targeted campaigns that focus on converting high-engagement casual riders into annual members. This includes promoting leisure-oriented membership benefits, introducing limited-time promotional offers or trial memberships, and scheduling campaigns during weekends and peak riding seasons when casual rider activity is highest. Marketing messages should emphasize the value of membership for frequent recreational riders rather than focusing exclusively on commuting benefits.

Following implementation, the marketing team should continuously monitor campaign performance using metrics such as membership conversion rates, campaign engagement, customer acquisition costs, and member retention. Regular evaluation of these metrics will enable the team to refine marketing strategies, optimize promotional efforts, and ensure that future campaigns remain aligned with rider behaviour and Cyclistic's objective of increasing annual memberships.

- What actions should the executive team take?

The executive team should use the findings from this analysis to guide strategic decisions related to membership growth, marketing investment, and product development. This includes prioritising resources toward data-driven marketing initiatives, supporting the introduction of flexible membership options that address the needs of casual riders, and encouraging cross-functional collaboration between marketing, operations, and customer experience teams to implement the recommended strategies effectively.

In addition, the executive team should establish clear performance objectives and regularly review key business metrics, including annual membership growth, conversion rates, customer acquisition costs, and member retention. Monitoring these outcomes will enable leadership to evaluate the effectiveness of the implemented strategies, make informed adjustments when necessary, and ensure that organisational efforts remain aligned with Cyclistic's long-term objective of increasing annual memberships.

- How could the success of these recommendations be measured?

The success of these recommendations can be measured by monitoring key performance indicators that reflect the effectiveness of Cyclistic's membership growth strategy. Primary metrics should include the conversion rate of casual riders to annual members, overall annual membership growth, customer acquisition costs, member retention rates, and the return on investment (ROI) of marketing campaigns. Tracking these indicators over time enables Cyclistic to determine whether the recommended strategies are successfully increasing membership adoption while maintaining cost-effective marketing performance.

In addition, Cyclistic should compare these metrics before and after implementing the recommended initiatives and continuously evaluate campaign performance during peak riding seasons. Regular performance reviews will help identify which strategies generate the highest conversion rates, allowing the company to refine future marketing efforts and support sustainable long-term membership growth.

Future Opportunities

- What additional data could strengthen future analyses?

Future analyses could be strengthened by incorporating additional customer and operational data that provides deeper insight into rider behaviour and membership conversion. Information such as rider demographics, membership purchase history, trip purpose, marketing campaign interactions, customer feedback, and promotional offer redemption would enable more detailed segmentation and help identify the factors that most strongly influence membership decisions. Geographic information, station capacity data, weather conditions, and special event schedules could also improve the understanding of external factors affecting ridership patterns.

Integrating these additional data sources would support more comprehensive behavioural analysis, enable more precise targeting of potential members, and improve the evaluation of marketing effectiveness. Access to richer customer and operational data would allow Cyclistic to develop more personalised, evidence-based strategies for increasing annual memberships and enhancing the overall customer experience.

- What future analyses could build upon this project?

Future analyses could build upon this project by investigating the long-term effectiveness of membership conversion strategies and evaluating how rider behaviour changes after casual riders become annual members. Comparative studies across multiple years could identify evolving seasonal trends, changes in rider preferences, and the sustained impact of marketing initiatives on membership growth. Additional analyses could also examine station-level demand, geographic riding patterns, and customer segmentation to support more targeted operational and marketing decisions.

Furthermore, predictive analytics and machine learning models could be developed to identify casual riders with the highest likelihood of purchasing an annual membership based on their historical riding behaviour.

These advanced analytical approaches would enable Cyclistic to proactively target high-potential customers, optimise marketing resource allocation, and further enhance data-driven decision-making for future membership growth initiatives.

- What limitations should stakeholders consider when implementing these recommendations?

Stakeholders should recognise that these recommendations are based on historical trip data and behavioural patterns observed during the analysis period. Rider preferences, travel demand, market conditions, and external factors such as weather, local events, infrastructure changes, or economic conditions may change over time, potentially influencing the effectiveness of the recommended strategies. Additionally, the analysis was based primarily on trip-level operational data and did not include customer demographics, marketing interactions, or qualitative feedback that could provide deeper insight into the motivations behind membership decisions.

Consequently, the recommendations should be viewed as evidence-based strategic guidance rather than guaranteed outcomes. Cyclistic should implement these initiatives alongside continuous performance monitoring and regularly evaluate key business metrics to assess their effectiveness. Ongoing data collection and periodic reanalysis will enable the company to refine its strategies, adapt to changing rider behaviour, and support sustained growth in annual memberships.